

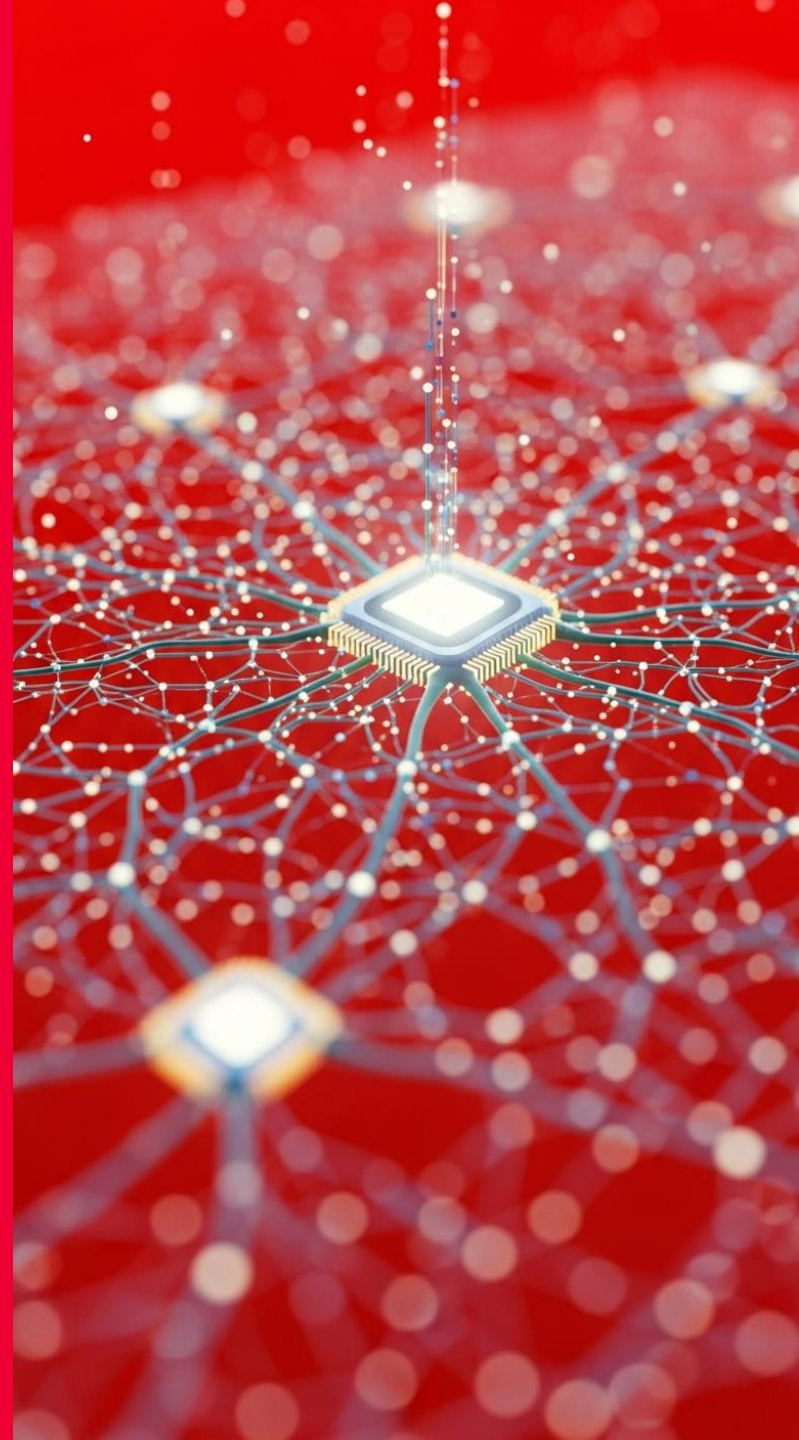
Cambridge Summer School

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# Artificial Intelligence fundamentals

Dr Mariano Felice

5<sup>th</sup> August 2026



# Outline

## **What is AI?**

- Definition
- Brief history

## **Types of AI**

- Symbolic methods
- Data-driven methods

## **Machine Learning**

- Definition
- Typical pipeline
- Supervised vs. unsupervised learning

## **Supervised Learning**

- Classification
- Regression

## **Evaluation**

- Evaluation metrics
- Overfitting and underfitting

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# **The origins of AI (1950–1960)**

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# Alan Turing: The Father of AI



Alan Turing  
1912 - 1954

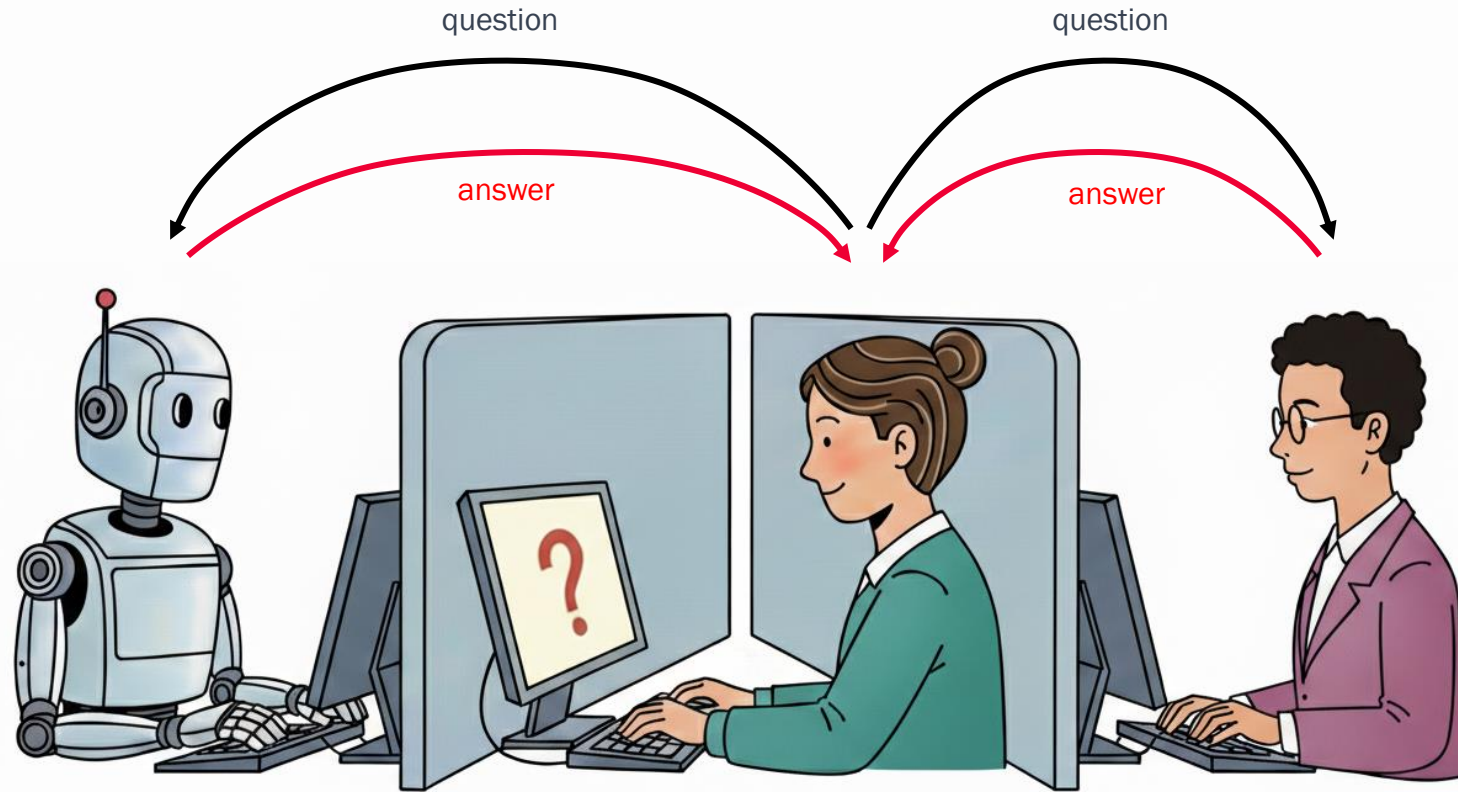
mathematician, computer pioneer  
and code breaker

Turing's seminal paper "Computing Machinery and Intelligence" (1950).

Intelligence is measured through the ability to talk like a human.

If a machine can perform every intellectual task a human can, we have no logical right to deny it "intelligence".

# The Imitation Game (a.k.a. Turing Test)



A person talks with both a human and a machine. If the person can't tell which is which, the machine is seen as intelligent.

# Over to you!

Which of these texts was generated by AI?

**A**

Society has a responsibility to take care of the environment. We all live on this planet, so it's important for us to do our part to keep it clean and healthy. There are many things we can do to be environmentally responsible. We should recycle our waste, such as paper, plastic, and cans, instead of throwing them away. It's also important to save energy by turning off lights and appliances when we're not using them. Using public transportation or carpooling can help reduce pollution from cars. We should also avoid wasting water and try to conserve it. Planting trees and taking part in community clean-up activities are great ways to contribute to a cleaner environment. By being environmentally responsible, we can make a positive impact on our planet and ensure a better future for ourselves and future generations.

**B**

Every country in the world suffers environmental catastrophes. This is a natural consequence of the struggle between development and environment. If a country decided to live isolated from the rest of the world, living on what it can naturally grow and produce, it surely wouldn't be highly polluted. But we all want exotic food and technological items from all over the world, so we have to pay the price. Investing on electrical transport would benefit the environment a lot. Even more if this electricity came from a natural source of energy like wind, rivers and solar boards. We also have to take care of our rivers and seas. We all have heard about factories throwing highly toxic substances to rivers, without minimizing their poisoning effects. A really strict law should be applied to fine these factories and make them change their policy.

# Over to you!

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**AI**

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**human**

**Over to you!**

Turing said that if a machine talks like a human, we should treat it as intelligent. Do you agree, or is there something “extra” inside us that a machine can never copy?

Does ChatGPT pass the Turing Test?

# The Dartmouth Workshop (1956)



A group of young scientists met at Dartmouth College (USA) to have a 2-month brainstorming session “to *find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves*”.

John McCarthy coined the term “Artificial Intelligence”.

## Over to you!

The scientists who took part in the Dartmouth Workshop were a group of men from one specific background (mostly US/UK mathematicians).

If you were going to hold a “Summer Workshop” today to define the future of AI, who would you invite and why?

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# What is AI?

Artificial Intelligence is the ability of computers to perform tasks that typically require human intelligence.

These are tasks that involve processes such as learning, thinking, understanding, reasoning, problem-solving, decision-making, inference, identifying objects, generating language, etc.

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# **The rule-based era (1960–1990)**

# Symbolic AI

Solutions were designed using clearly defined algorithms, heuristics, logical representations or IF-THEN rules.

Examples: rule-based systems and search algorithms.

The knowledge is coded in the *algorithm*.

```
<rule>
  <pattern>
    <token regexp="yes">what|which</token>
    <token postag="JJ.*|NN.*" postag_regexp="yes" min="0" />
    <token postag="NN.*" postag_regexp="yes" chunk_re="E-NP.*" />
    <token postag="MD">
      <exception>need</exception>
    </token>
    <token inflected="yes" min="0">not</token>
    <token
  regexp="yes">s?he|it|I|we|they|you|this|that|th[oe]se| (some|any|every|no) (thing|bo
  dy|one)</token>
    <token postag="RB" min="0">
      <exception postag="IN|EX" postag_regexp="yes" />
      <exception regexp="yes">live|close|last</exception>
    </token>
    <marker>
      <token postag="VB[GDNZ]" postag_regexp="yes" skip="-1">
        <exception postag="VB" />
        <exception regexp="yes">kids|physics|knights|let|fit</exception>
        <exception>'s</exception>
      </token>
    </marker>
    <token>?</token>
  </pattern>
  <message>The correct verb form after the modal verb "<match
  no="8" postag="VB" /></suggestion>.</message>
  <example correction="buy">Which car would you
  <marker>bought</marker>?</example>
  <example correction="choose">Which one would you <marker>chose</marker> and
  why?</example>
</rule>
```

A grammar rule from LanguageTool.

\_\_\_\_\_

ELIZA was a simple program created at the MIT that mimicked a therapist.

It worked by flipping the user's questions back at them using predefined rules.

Welcome to

```

EEEEEE LL      IIII  ZZZZZZ  AAAAA
EE      LL      II    ZZ      AA   AA
EEEEEE LL      II    ZZZ      AAAAAA
EE      LL      II    ZZ      AA   AA
EEEEEE LLLLLL  IIII  ZZZZZZ  AA   AA

```

Eliza is a mock Rogerian psychotherapist.  
The original program was described by Joseph Weizenbaum in 1966.  
This implementation by Norbert Landsteiner 2005.

```
ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

|  | PRINT,TQ109,2531,,TAPE,,102                                           | TQ109 2531 |
|--|-----------------------------------------------------------------------|------------|
|  | (HOW DO YOU DO. I AM THE DOCTOR. PLEASE SIT DOWN AT THE TYPEWRITER    | 000010     |
|  | AND TELL ME YOUR PROBLEM.)                                            | 000020     |
|  | (IF 3 ((O IF O) (DO YOU THINK ITS LIKELY THAT 3) (DO YOU WISH THAT 3) | 000030     |
|  | (WHAT DO YOU THINK ABOUT 3) (REALLY, 2 3)))                           | 000040     |
|  | (MEMORY MY (O YOUR O = LETS DISCUSS FURTHER WHY YOUR 3)               | 000050     |
|  | (O YOUR O = EARLIER YOU SAID YOUR 3)                                  | 000060     |
|  | (O YOUR O = BUT YOUR 3)                                               | 000070     |
|  | (O YOUR O = DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR 3)) | 000080     |
|  | (NONE ((O) (I AM NOT SURE I UNDERSTAND YOU FULLY)                     | 000090     |
|  | (PLEASE GO ON)                                                        | 000100     |
|  | (WHAT DOES THAT SUGGEST TO YOU)                                       | 000110     |
|  | (DO YOU FEEL STRONGLY ABOUT DISCUSSING SUCH THINGS)))                 | 000120     |
|  | (PERHAPS ((O) (YOU DON'T SEEM QUITE CERTAIN)                          | 000130     |
|  | (WHY THE UNCERTAIN TONE)                                              | 000140     |
|  | (CAN'T YOU BE MORE POSITIVE)                                          | 000150     |
|  | (YOU AREN'T SURE)))                                                   | 000180     |
|  | (MAYBE (=PERHAPS))                                                    | 000190     |
|  | (DON'T YOU KNOW)                                                      | 000200     |
|  | (AM = ARE ((O ARE YOU O) (DO YOU BELIEVE YOU ARE 4)                   | 000210     |
|  | (WOULD YOU WANT TO BE 4) (YOU WISH I WOULD TELL YOU YOU ARE 4)        | 000220     |
|  | (WHAT WOULD IT MEAN IF YOU WERE 4))                                   | 000230     |
|  | ((O) (WHY DO YOU SAY 'AM') (I DON'T UNDERSTAND THAT)))                | 000240     |
|  | (ARE = AM ((O AM I O)                                                 | 000250     |
|  | (WHY ARE YOU INTERESTED IN WHETHER I AM 4 OR NOT)                     | 000260     |
|  | (WOULD YOU PREFER IF I WEREN'T 4) (PERHAPS I AM 4 IN YOUR             | 000270     |
|  | FANTASIES) (DO YOU SOMETIMES THINK I AM 4))                           | 000280     |
|  | ((O AM O) (DID YOU THINK THEY MIGHT NOT BE 3)                         | 000290     |
|  | (WOULD YOU LIKE IT IF THEY WERE NOT 3) (WHAT IF THEY WERE NOT 3)      | 000300     |
|  | (POSSIBLY THEY ARE 3))                                                | 000310     |
|  | (YOUR = MY ((O MY O) (WHY ARE YOU CONCERNED OVER MY 3)                | 000320     |
|  | (WHAT ABOUT YOUR OWN 3) (ARE YOU WORRIED ABOUT SOMEONE ELSE'S 3)      | 000330     |
|  | (REALLY, MY 3)))                                                      | 000340     |
|  | (WAS = WERE)                                                          | 000350     |
|  | (WERE = WAS)                                                          | 000360     |
|  | (ME = YOU)                                                            | 000370     |
|  | (YOU'RE = I'M)                                                        | 000380     |
|  | (I'M = YOU'RE)                                                        | 000390     |
|  | (MYSELF = YOURSELF)                                                   | 000400     |
|  | (YOURSELF = MYSELF)                                                   | 000410     |
|  | (MOTHER DLIST(/NOUN FAMILY))                                          | 000420     |
|  | (FATHER DLIST(/NOUN FAMILY))                                          | 000430     |
|  | (SISTER DLIST(/FAMILY))                                               | 000440     |
|  | (BROTHER DLIST(/FAMILY))                                              | 000450     |
|  | (WIFE DLIST(/FAMILY))                                                 | 000460     |
|  | (CHILDREN DLIST(/FAMILY))                                             | 000470     |
|  | (I = YOU ((O YOU ARE O I O)                                           | 000480     |

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## The AI Winter (late 1970s)

AI faced strong criticism, as many scientists believed AI research had little real-world value and was something impossible to achieve.

Computers were too slow, rule-based were too rigid and couldn't handle the messiness of the real world.

As many projects failed to deliver their promises, funding for AI research dried up.



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## Expert Systems (1980s)

*Expert systems* were big rule-based systems that attempted to mimic the human decision-making and reasoning of experts in a particular domain, e.g. medicine.

To build them, experts had to explain their thinking step by step. This process, called knowledge engineering, was laborious and time-consuming. The systems couldn't handle new situations or learn from their mistakes.

**Over to you!**

Can you speak a new language only by reading a dictionary and a grammar book? Why/why not?

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# More Flexible Approaches

Researchers explored methods that could handle uncertainty and learn from experience.

Examples:

- Fuzzy logic
- Probabilistic methods
- Genetic algorithms
- Neural Networks

There was a shift towards statistical methods.

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# **The data age (1990–2010)**

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# Data-driven AI: Machine Learning (ML)

Machines can “learn” from examples instead of descriptions.  
E.g. don't tell the machine what a cat is; show it a million cats.

The success of ML was due to the increasing availability of big volumes of (training) data.

The knowledge is in the *data* (we don't need domain experts any more).

ML can solve virtually any problem (and more accurately than before).

# Machine Learning Algorithms

Some well-known algorithms:

- Logistic Regression
- Neural Networks
- Support Vector Machines
- Random Forests
- Conditional Random Fields
- K-Nearest Neighbours

Some tasks that use ML:

- Automatic Speech Recognition
- Text to speech
- Machine translation
- Sentiment analysis
- Image recognition
- Recommendation systems
- ...

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# **The power of Deep Learning (2010–2017)**

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# Deep Learning

- Big neural networks, with more than two hidden layers.
- Adding more layers increases their expressive power but also requires massive volumes of data and more powerful computational resources (GPUs).
- These networks can identify what is important in the input all by themselves – no need to explicitly define “features”.
- Many of these models are “end-to-end”, i.e. they can learn to predict values from the input without explicit intermediate representations.
- Machines gain the ability to “understand” meaning.
- Greater abstraction capabilities and performance.
- Huge advances in tasks such as machine translation, image processing, summarisation, etc.

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# **The Generative AI revolution (2017–today)**

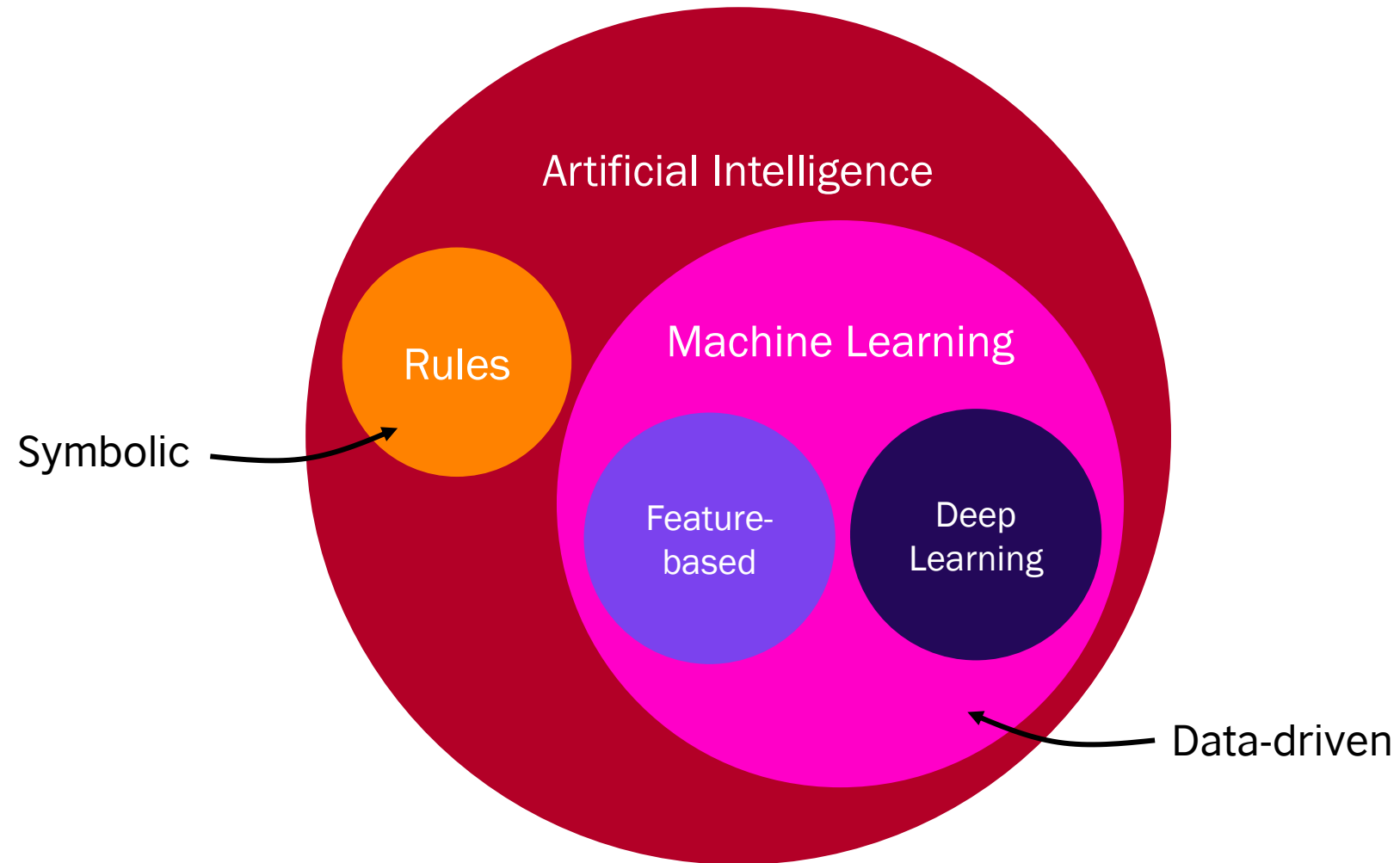
# Generative AI



- The emergence of new powerful neural networks (known as “Transformers”) gave machines greater understanding capabilities.
- Large Language Models such as ChatGPT (2022) revolutionised the field and led to an unprecedented interest and “democratisation” of AI.
- AI moved from “recognition” to “generation”.
- Anyone can generate text, images, video, sound, etc. with just a “prompt” (plain text instructions) – no need to code or be an AI expert.
- The quality of the results depend on the quality of the prompt (“prompt engineering”).
- Results are not perfect – they may contain hallucinations and inaccuracies.

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# Summary of AI approaches



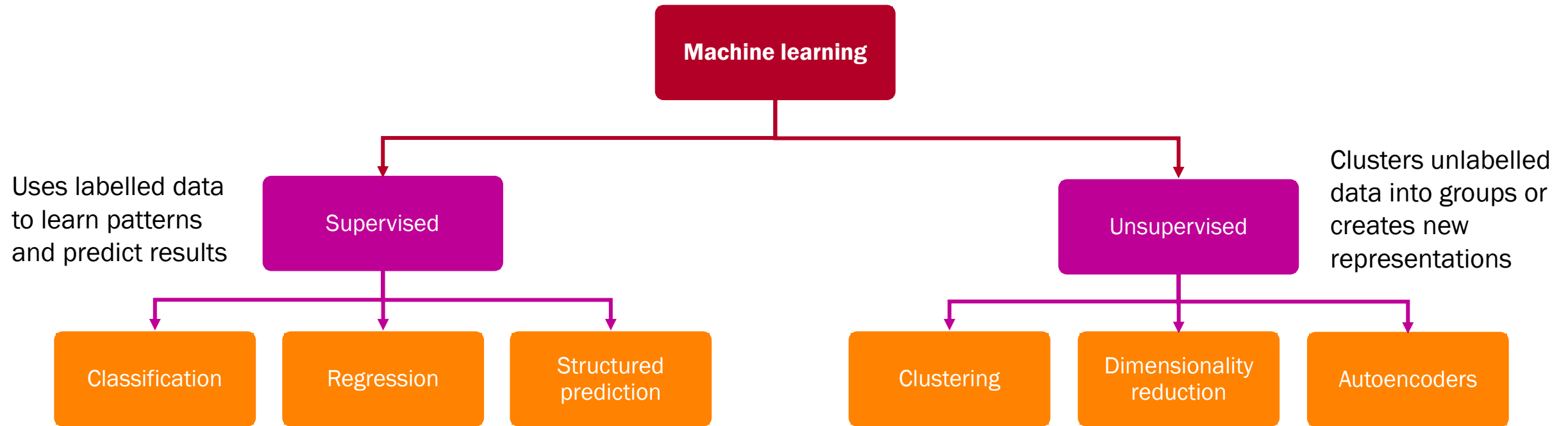


**How does modern AI work?**

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**How does modern AI work?**

**Machine Learning**



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# Supervised learning

## Regression

Learn to predict a numerical value (discrete or continuous).

### EXAMPLES

- Score a piece of writing automatically.
- Predict stock prices.
- Predict temperature.
- Estimate sales.

---

# Supervised learning

## Classification

- Assign a class to an instance/data point.
- Classification can be binary or multi-class.

### EXAMPLES

#### Binary

- Spam detection.
- Pass/fail an exam.
- Test positive/negative for a health condition.

#### Multiclass

- Image classification.
- Classify students by language proficiency level.
- Classify dogs into breeds.

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# Supervised learning

## Structured prediction

Predict structured output, i.e. sequences.

### EXAMPLES

- Part-of-speech tagging.

This is a nice house  $\Rightarrow$  This/PRON is/VERB a/DET nice/ADJ house/NOUN

- Optical character recognition.

My car is sick  $\Rightarrow$  My cat is sick

- Machine translation.

Il libro è interessante.  $\Rightarrow$  The book is interesting.

---

# Unsupervised learning

## Clustering

Group instances into classes based on their similarity.

### EXAMPLES

- Group customers by purchasing habits.
- Group articles by topics.
- Segment areas based on criminal activity.

# Unsupervised learning

## Dimensionality reduction

Reduce a large collection of values, such as a vector of features, into a more compact representation containing the most salient features.

### EXAMPLE

A vector containing 1,000 values that represent different aspects of a word (length, absolute frequency, relative frequency, POS, number of times it follows a preposition, number of times it follows a noun, etc.)

[3.63, 657, 269.1, 63.36, 8, 0.4536, 13.5, ..., 245.134, 25.17, 0.9, 6.72, 0.92, 0.0001, 52]



[0.3, 0.536, 3223, 335.8, 1, 145, 83.6, ..., 829.5]

A new vector of 100 dimensions that contains the most important information.

---

# Unsupervised learning

## Autoencoders

The representation of an instance as a vector of numerical values.

### EXAMPLE

Representing a sentence as a vector of numbers.

Monkeys are agile,  
intelligent primates.     →     [25.32, 6.2, 2.534, 0.23, 0.004, 682, ..., 0.53, 3.758]

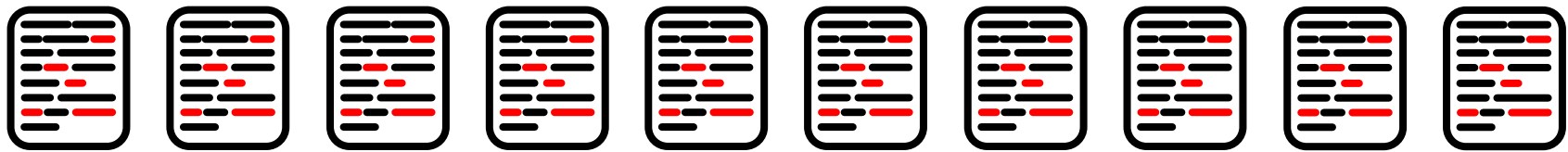


**How does supervised machine learning work?**



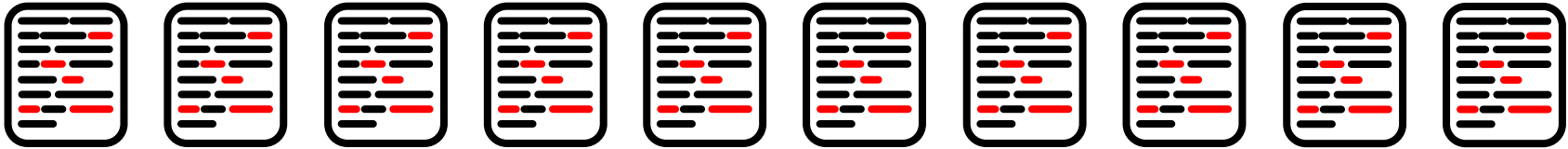
# Data collection

Examples



# Data collection

Examples

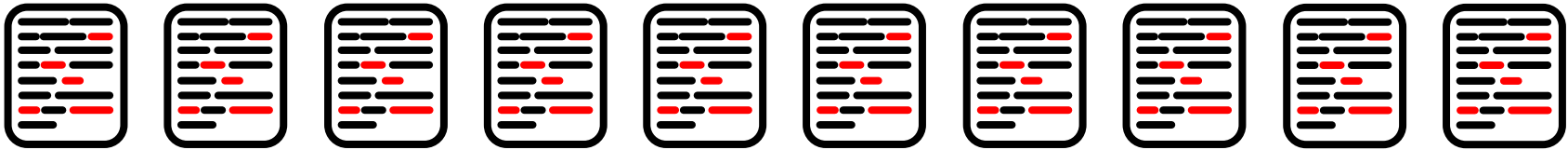


Labels

2 3 8 2 9 3 7 7 10 4

# Data collection

Examples

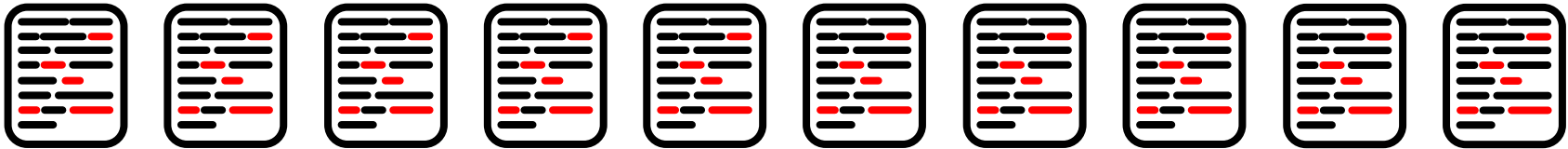


Labels

GOOD BAD GOOD BAD BAD GOOD GOOD GOOD GOOD BAD

# Data collection

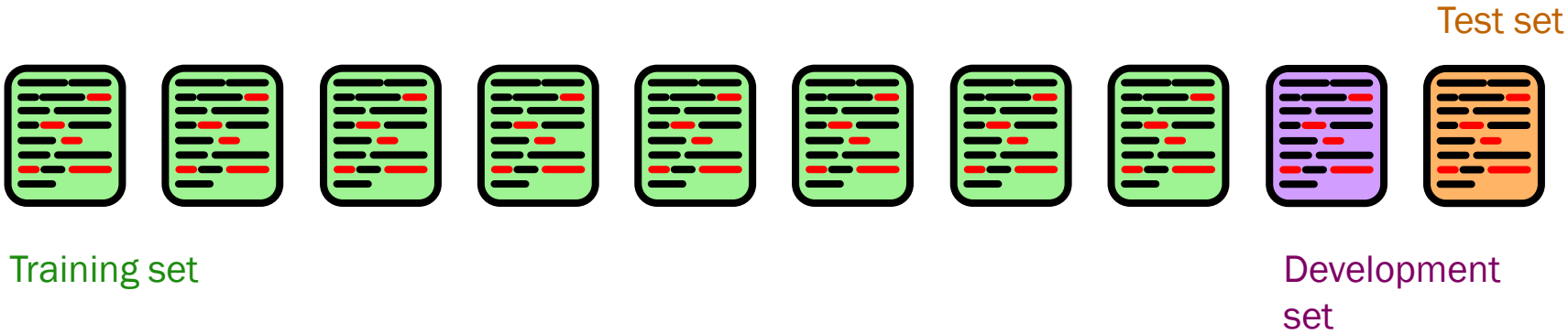
Examples



Labels

EASY MEDIUM MEDIUM EASY HARD EASY HARD HARD HARD MEDIUM

# Data collection



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# Training

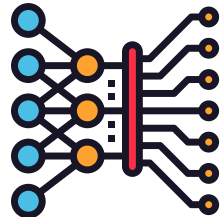


Training  
data

# Training



Training  
data

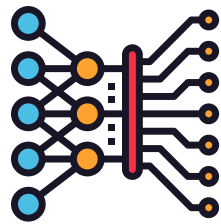


Machine  
learning

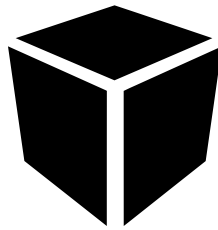
# Training



Training  
data



Machine  
learning

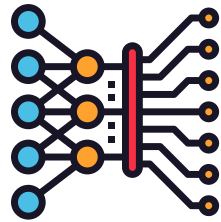


Model

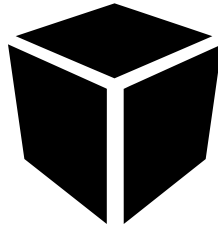
# Fine-tuning



Training  
data

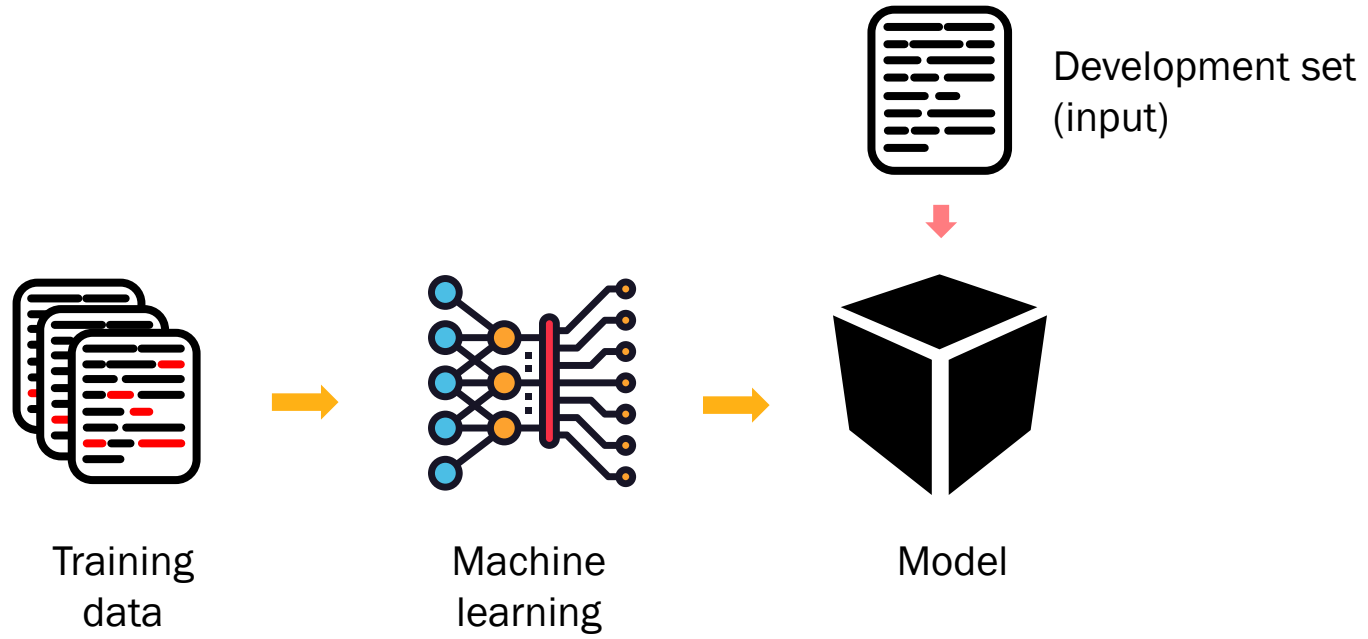


Machine  
learning

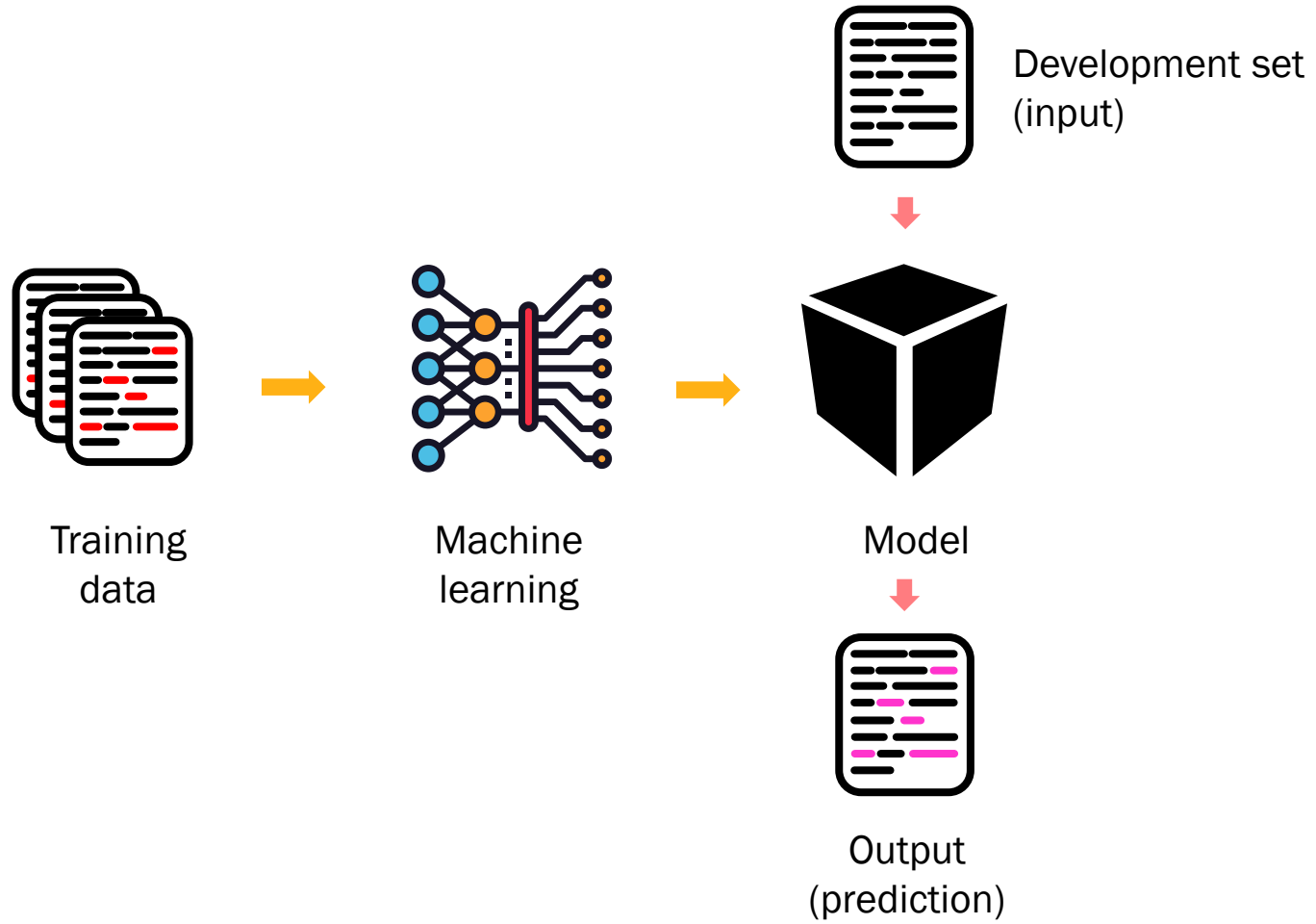


Model

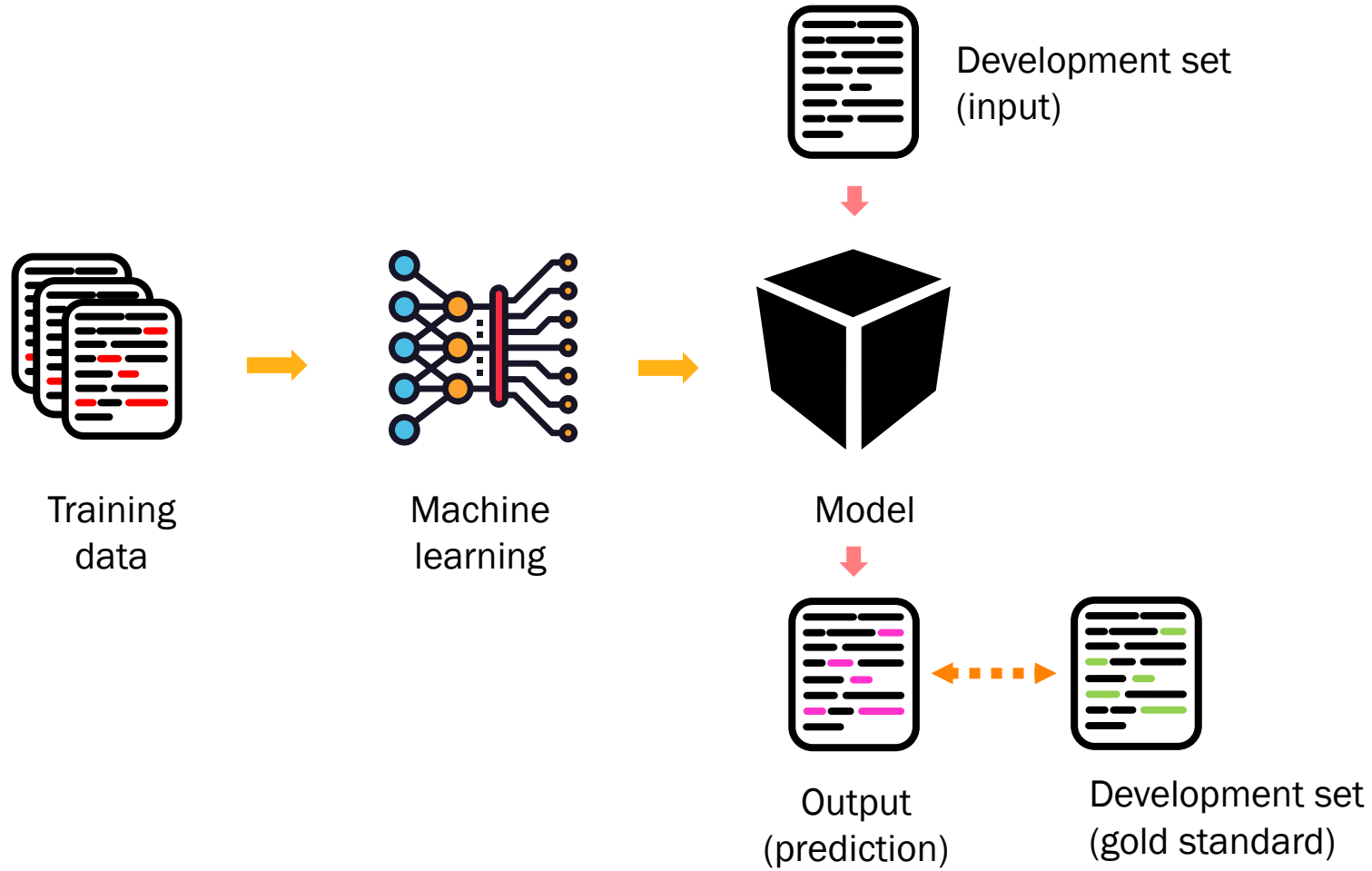
# Fine-tuning



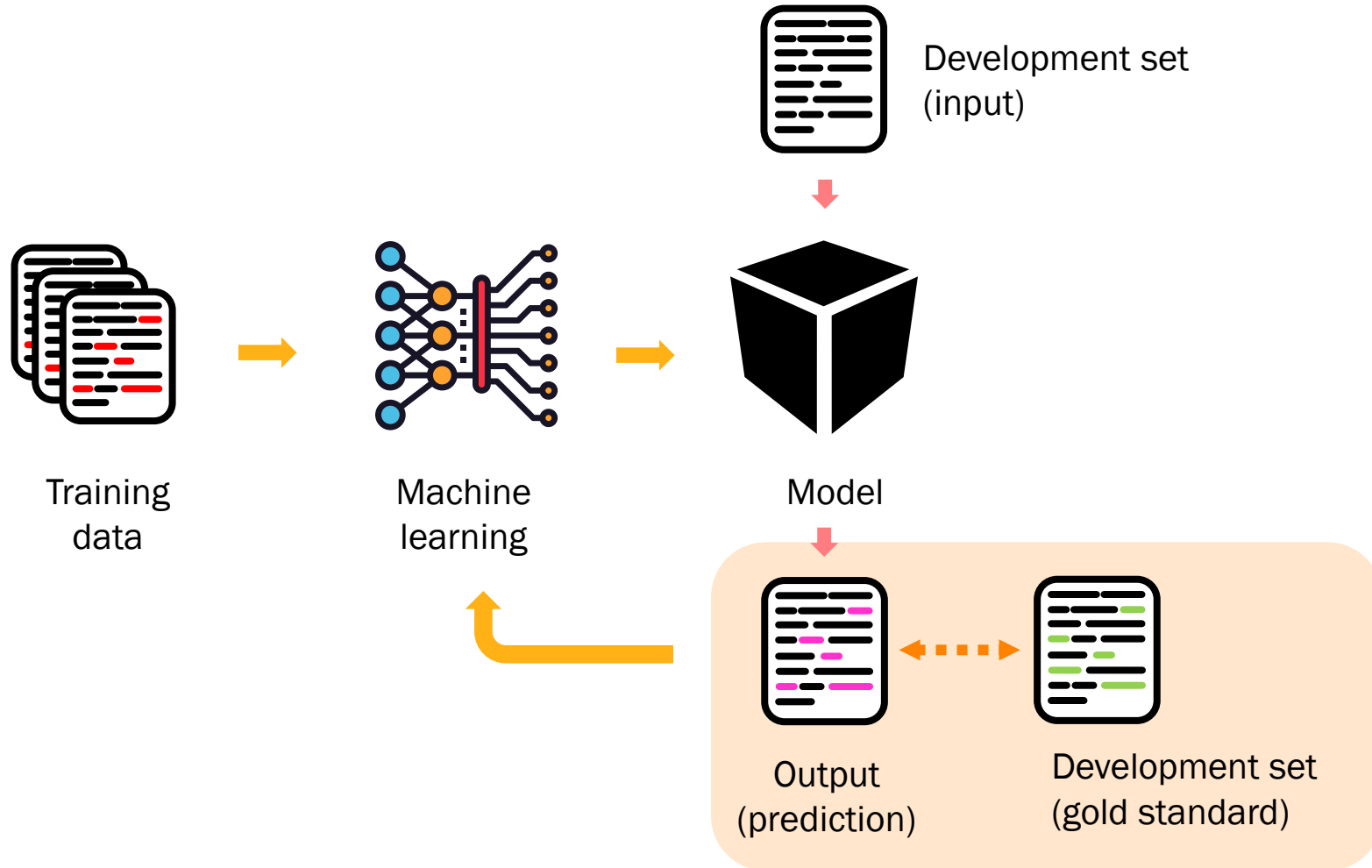
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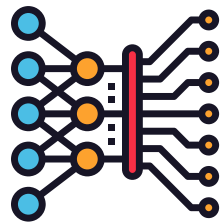
# Fine-tuning



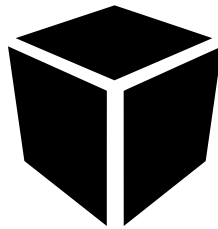
# Testing



Training  
data

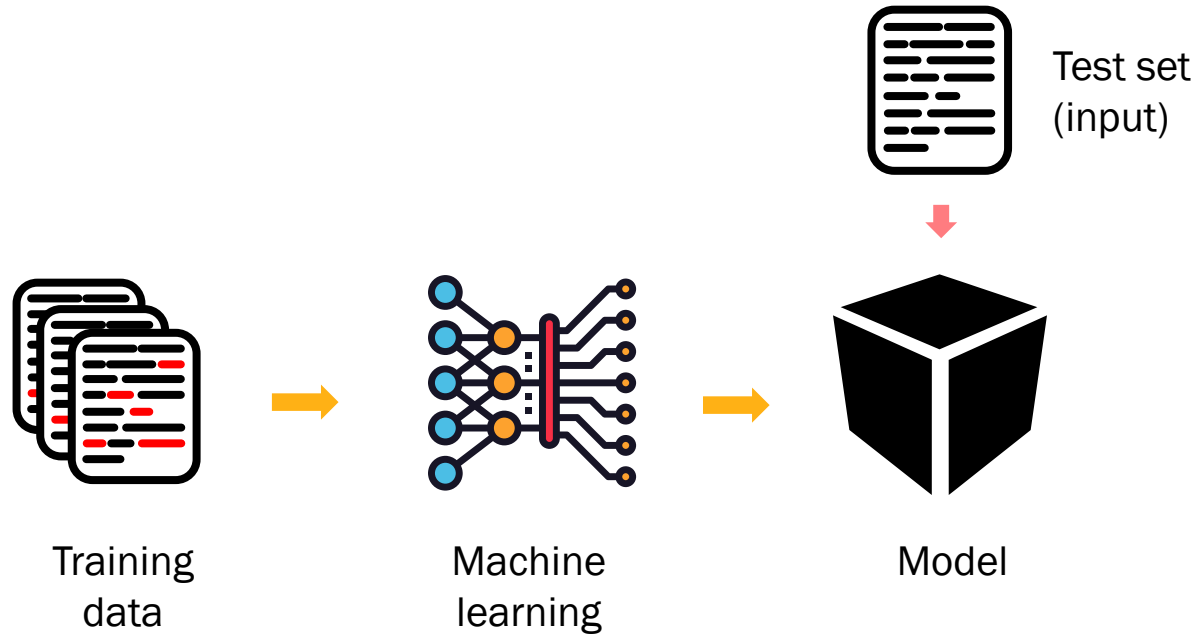


Machine  
learning

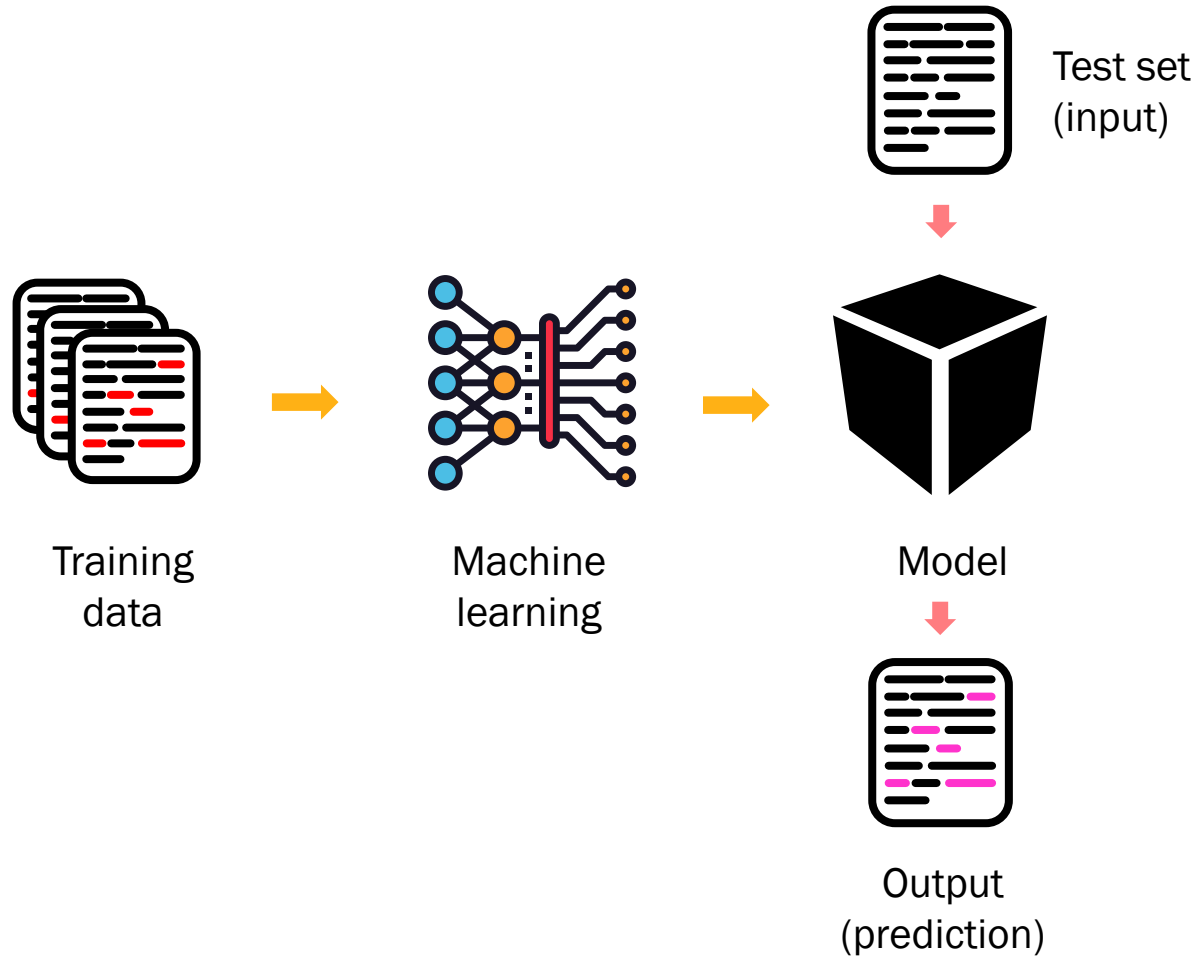


Model

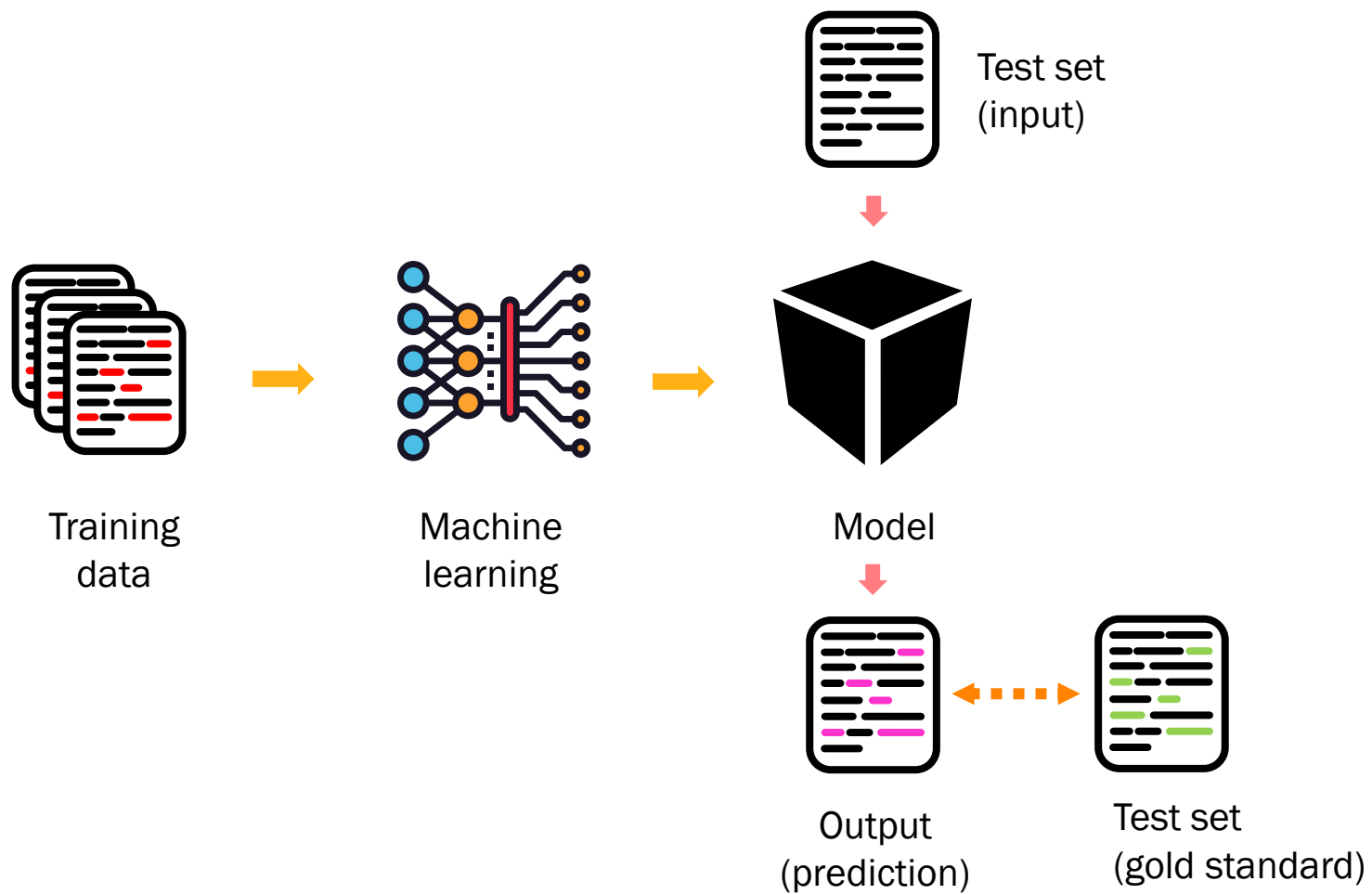
# Testing



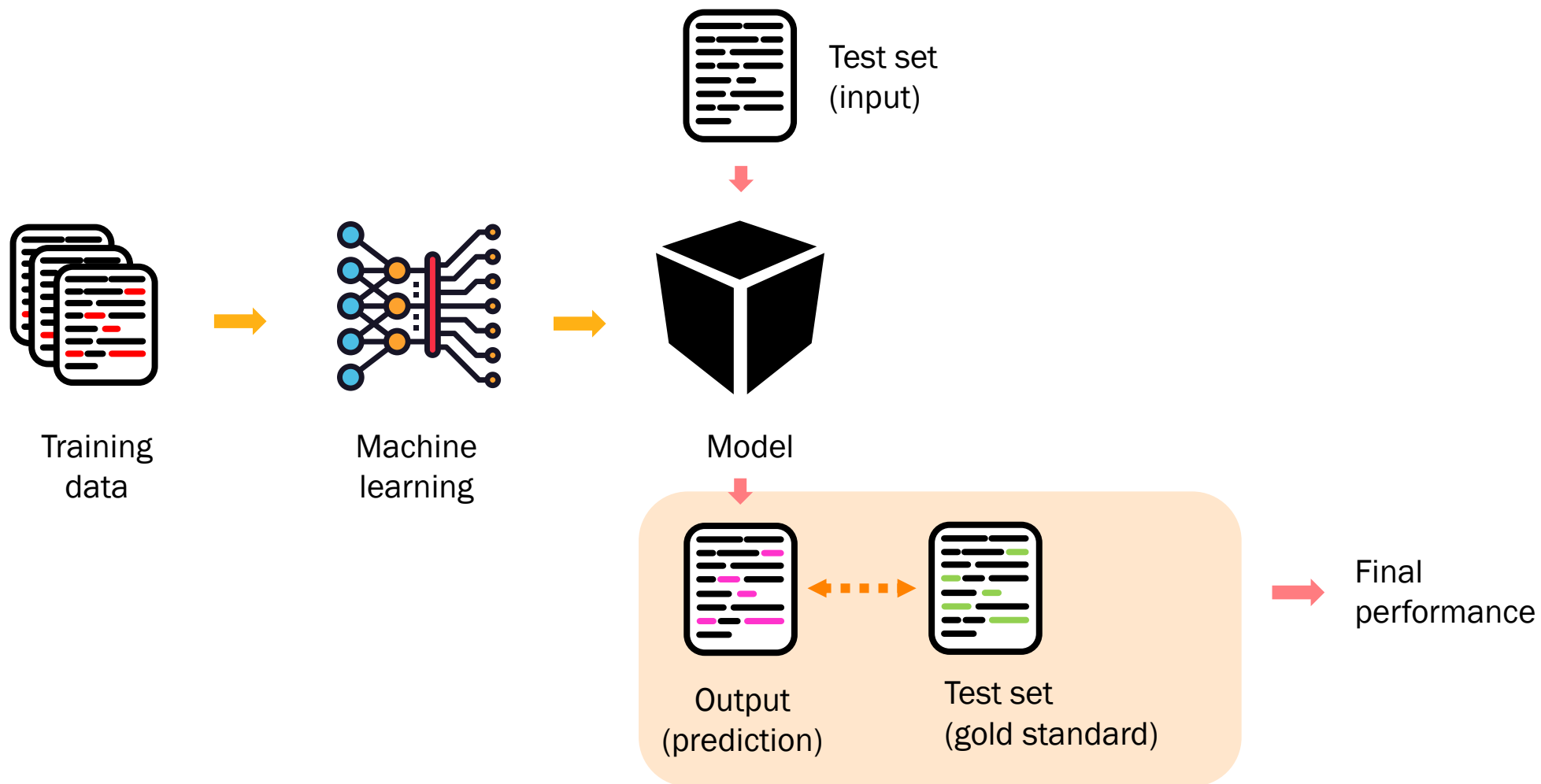
# Testing



# Testing



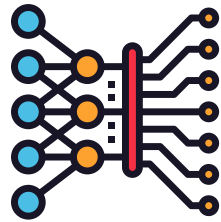
# Testing



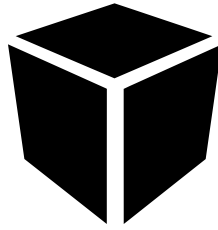
# Real-world usage



Training  
data

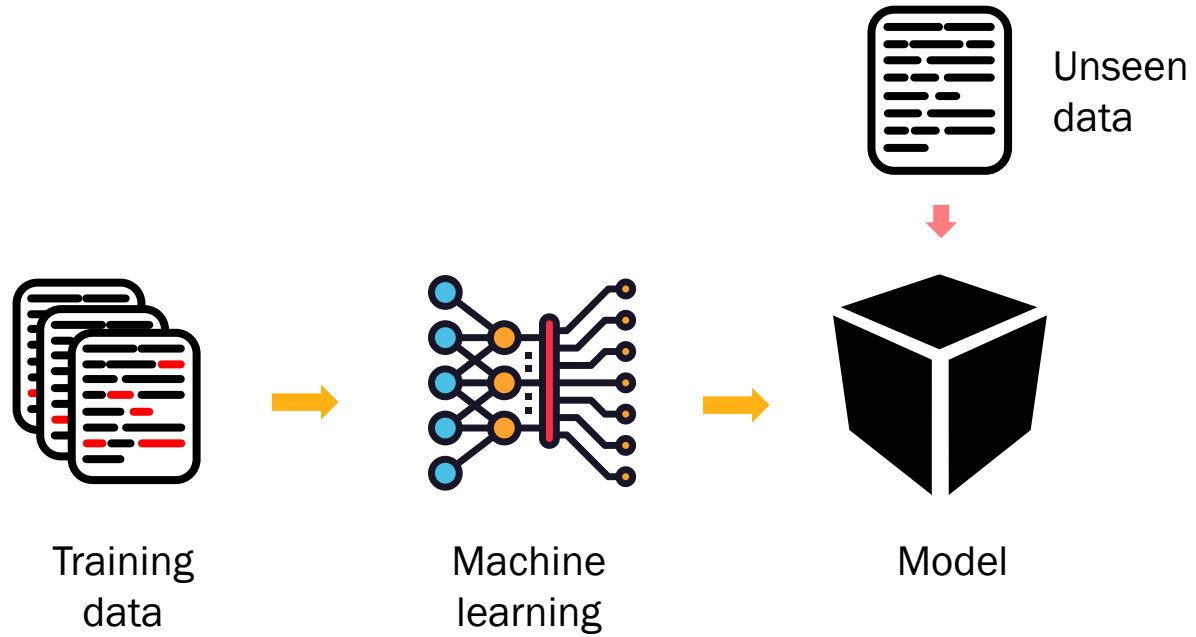


Machine  
learning

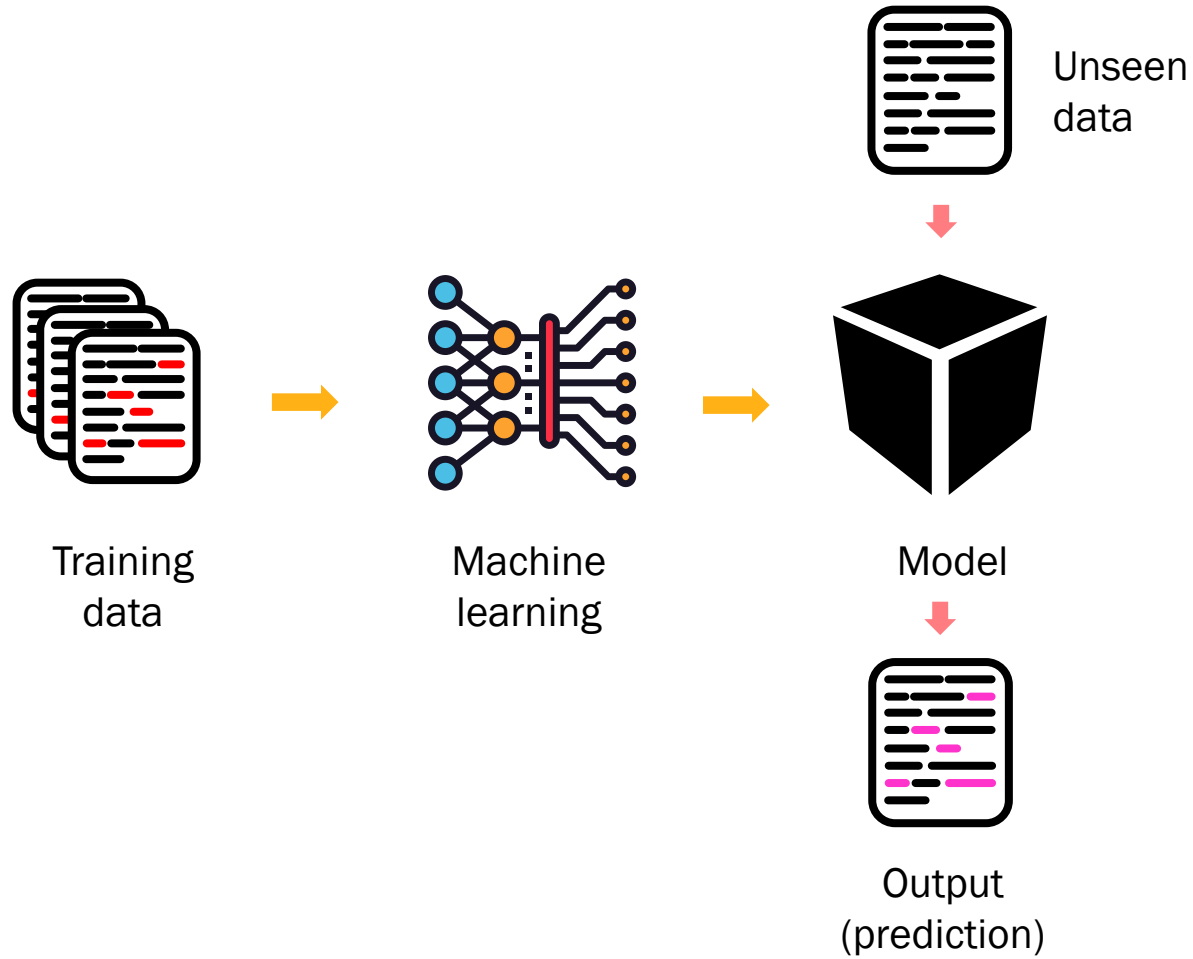


Model

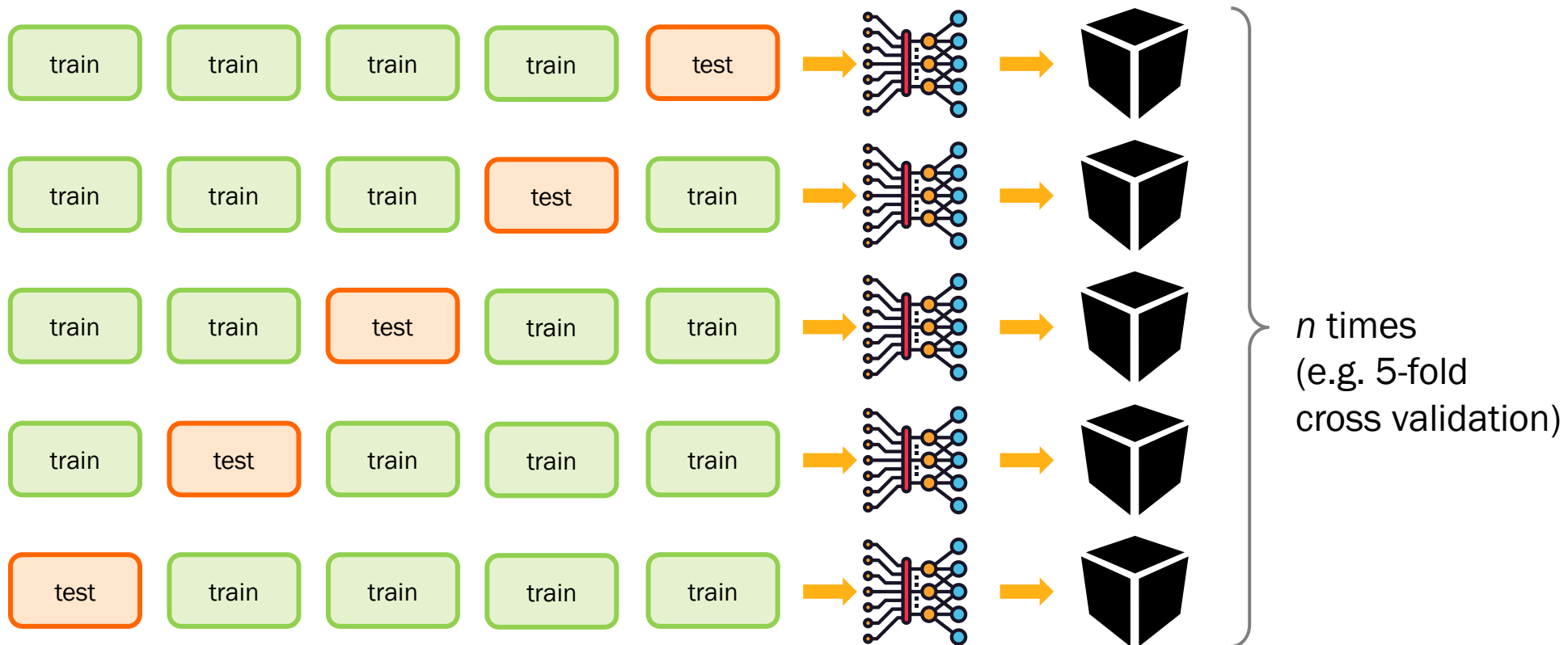
# Real-world usage



# Real-world usage



# Cross validation



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# Feature-based models

Training instances are defined as a set of “features”, i.e. information about different aspects of the input that can help us predict the target value.

**Example:** Determine if a patient has COVID, flu, a cold or an allergy based on their symptoms.

What information about each patient and their symptoms would be useful?

# Feature-based models

Training instances are defined as a set of “features”, i.e. information about different aspects of the input that can help us predict the target value.

**Example:** Determine if a patient has COVID, flu, a cold or an allergy based on their symptoms.

What information about each patient and their symptoms would be useful?

**Age:** Patient age (18 to 85 years)

**Sex:** Biological sex (Female, Male)

**Preexisting Condition:** Medical background (None, Asthma, Diabetes)

**Primary Symptom:** Chief complaint (Sneezing, Runny Nose, Dry Cough, Shortness of Breath)

**Temperature C:** Body temperature (36.0 to 40.0 °C)

**CRP mgL:** Inflammatory marker (0.0 to 100.0+ mg/L)

**SpO2 Percent:** Blood oxygen (85.0 to 100.0 %)

**Eosinophils Percent:** Allergy biomarker (0.0 to 10.0 %)

**Diagnosis:** Target class (Allergy, Common Cold, COVID19, Influenza)

# Our dataset

| Age | Sex    | Preexisting_Condition | Primary_Symptom     | Temperature_C | CRP_mgL | SpO2_Percent | Eosinophils_Percent | Diagnosis   |
|-----|--------|-----------------------|---------------------|---------------|---------|--------------|---------------------|-------------|
| 38  | Female | None                  | Sneezing            | 36.7          | 0.8     | 98.6         | 7.1                 | Allergy     |
| 52  | Male   | Asthma                | Dry_Cough           | 38.3          | 62.4    | 93.1         | 0.9                 | COVID19     |
| 29  | Female | None                  | Runny_Nose          |               | 11.5    | 97.8         | 2.1                 | Common_Cold |
| 61  | Male   | Diabetes              | Dry_Cough           | 38.9          | 48.2    | 95.2         | 1.1                 | Influenza   |
| 44  | Female | None                  | Sneezing            | 36.9          | 1.2     | 98.4         | 6.2                 | Allergy     |
| 33  | Male   | None                  | Shortness_of_Breath | 38.5          | 58.1    | 92.4         | 0.8                 | COVID19     |
| 57  | Female | Asthma                | Runny_Nose          | 37.6          | 13.8    | 97.2         | 1.8                 | Common_Cold |
| 48  | Male   | None                  | Dry_Cough           | 999           | 999     | 96.1         | 1.3                 | Influenza   |
| 22  | Female | None                  | Sneezing            |               | 0.5     | 99.1         | 6.8                 | Allergy     |
| 65  | Male   | Diabetes              | Shortness_of_Breath | 38.2          | 68.3    | 91.8         | 1.0                 | COVID19     |
| 41  | Female | None                  | Runny_Nose          | 37.3          | 9.8     | 98.0         | 2.4                 | Common_Cold |
| 50  | Male   | Asthma                | Dry_Cough           | 38.4          | 49.5    | 95.8         | 1.2                 | Influenza   |

# Preprocessing

We can't use the data “as is” – we have to prepare it for machine learning.

| Age | Sex    | Preexisting_Condition | Primary_Symptom     | Temperature_C | CRP_mgL | SpO2_Percent | Eosinophils_Percent | Diagnosis   |
|-----|--------|-----------------------|---------------------|---------------|---------|--------------|---------------------|-------------|
| 38  | Female | None                  | Sneezing            | 36.7          | 0.8     | 98.6         | 7.1                 | Allergy     |
| 52  | Male   | Asthma                | Dry_Cough           | 38.3          | 62.4    | 93.1         | 0.9                 | COVID19     |
| 29  | Female | None                  | Runny_Nose          |               | 11.5    | 97.8         | 2.1                 | Common_Cold |
| 61  | Male   | Diabetes              | Dry_Cough           | 38.9          | 48.2    | 95.2         | 1.1                 | Influenza   |
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| 57  | Female | Asthma                | Runny_Nose          | 37.6          | 13.8    | 97.2         | 1.8                 | Common_Cold |
| 48  | Male   | None                  | Dry_Cough           | 999           | 999     | 96.1         | 1.3                 | Influenza   |
| 22  | Female | None                  | Sneezing            |               | 0.5     | 99.1         | 6.8                 | Allergy     |
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# Preprocessing

We may need to fix any problems in our data, such as missing values or corrupt records.

| Age | Sex    | Preexisting_Condition | Primary_Symptom     | Temperature_C | CRP_mgL | SpO2_Percent | Eosinophils_Percent | Diagnosis   |
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| 65  | Male   | Diabetes              | Shortness_of_Breath | 38.2          | 68.3    | 91.8         | 1.0                 | COVID19     |
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# Preprocessing

We may need to scale numerical ranges so that machine learning is more stable and converges more easily.

| Age | Sex    | Preexisting_Condition | Primary_Symptom     | Temperature_C | CRP_mgL | SpO2_Percent | Eosinophils_Percent | Diagnosis   |
|-----|--------|-----------------------|---------------------|---------------|---------|--------------|---------------------|-------------|
| 38  | Female | None                  | Sneezing            | 36.7          | 0.8     | 98.6         | 7.1                 | Allergy     |
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| 41  | Female | None                  | Runny_Nose          | 37.3          | 9.8     | 98.0         | 2.4                 | Common_Cold |
| 50  | Male   | Asthma                | Dry_Cough           | 38.4          | 49.5    | 95.8         | 1.2                 | Influenza   |

# Preprocessing

We may need to scale numerical ranges so that machine learning is more stable and converges more easily.

| Age    | Sex    | Preexisting_Condition | Primary_Symptom     | Temperature_C | CRP_mgL | SpO2_Percent | Eosinophils_Percent | Diagnosis   |
|--------|--------|-----------------------|---------------------|---------------|---------|--------------|---------------------|-------------|
| -0.455 | Female | None                  | Sneezing            | -1.332        | -1.066  | 0.999        | 1.488               | Allergy     |
| 0.639  | Male   | Asthma                | Dry_Cough           | 0.725         | 1.258   | -1.189       | -0.73               | COVID19     |
| -1.159 | Female | None                  | Runny_Nose          | -1.203        | -0.662  | 0.681        | -0.301              | Common_Cold |
| 1.343  | Male   | Diabetes              | Dry_Cough           | 1.496         | 0.722   | -0.354       | -0.658              | Influenza   |
| 0.014  | Female | None                  | Sneezing            | -1.075        | -1.051  | 0.919        | 1.166               | Allergy     |
| -0.846 | Male   | None                  | Shortness_of_Breath | 0.982         | 1.096   | -1.468       | -0.765              | COVID19     |
| 1.03   | Female | Asthma                | Runny_Nose          | -0.175        | -0.575  | 0.442        | -0.408              | Common_Cold |
| -1.706 | Female | None                  | Sneezing            | -1.203        | -1.077  | 1.198        | 1.38                | Allergy     |
| 1.656  | Male   | Diabetes              | Shortness_of_Breath | 0.597         | 1.481   | -1.706       | -0.694              | COVID19     |
| -0.22  | Female | None                  | Runny_Nose          | -0.561        | -0.726  | 0.76         | -0.194              | Common_Cold |
| 0.483  | Male   | Asthma                | Dry_Cough           | 0.854         | 0.771   | -0.115       | -0.623              | Influenza   |

# Preprocessing

Machines only understand numbers, so we must convert all our data to a numerical representation.

| Age    | Sex    | Preexisting_Condition | Primary_Symptom     | Temperature_C | CRP_mg/L | SpO2_Percent | Eosinophils_Percent | Diagnosis   |
|--------|--------|-----------------------|---------------------|---------------|----------|--------------|---------------------|-------------|
| -0.455 | Female | None                  | Sneezing            | -1.332        | -1.066   | 0.999        | 1.488               | Allergy     |
| 0.639  | Male   | Asthma                | Dry_Cough           | 0.725         | 1.258    | -1.189       | -0.73               | COVID19     |
| -1.159 | Female | None                  | Runny_Nose          | -1.203        | -0.662   | 0.681        | -0.301              | Common_Cold |
| 1.343  | Male   | Diabetes              | Dry_Cough           | 1.496         | 0.722    | -0.354       | -0.658              | Influenza   |
| 0.014  | Female | None                  | Sneezing            | -1.075        | -1.051   | 0.919        | 1.166               | Allergy     |
| -0.846 | Male   | None                  | Shortness_of_Breath | 0.982         | 1.096    | -1.468       | -0.765              | COVID19     |
| 1.03   | Female | Asthma                | Runny_Nose          | -0.175        | -0.575   | 0.442        | -0.408              | Common_Cold |
| -1.706 | Female | None                  | Sneezing            | -1.203        | -1.077   | 1.198        | 1.38                | Allergy     |
| 1.656  | Male   | Diabetes              | Shortness_of_Breath | 0.597         | 1.481    | -1.706       | -0.694              | COVID19     |
| -0.22  | Female | None                  | Runny_Nose          | -0.561        | -0.726   | 0.76         | -0.194              | Common_Cold |
| 0.483  | Male   | Asthma                | Dry_Cough           | 0.854         | 0.771    | -0.115       | -0.623              | Influenza   |

# Preprocessing

Machines only understand numbers, so we must convert all our data to a numerical representation.

| Age    | Sex | Preexisting_Condition | Primary_Symptom | Temperature_C | CRP_mgL | SpO2_Percent | Eosinophils_Percent | Diagnosis |
|--------|-----|-----------------------|-----------------|---------------|---------|--------------|---------------------|-----------|
| -0.455 | 0   | 0                     | 0               | -1.332        | -1.066  | 0.999        | 1.488               | 0         |
| 0.639  | 1   | 1                     | 1               | 0.725         | 1.258   | -1.189       | -0.73               | 1         |
| -1.159 | 0   | 0                     | 2               | -1.203        | -0.662  | 0.681        | -0.301              | 2         |
| 1.343  | 1   | 2                     | 1               | 1.496         | 0.722   | -0.354       | -0.658              | 3         |
| 0.014  | 0   | 0                     | 0               | -1.075        | -1.051  | 0.919        | 1.166               | 0         |
| -0.846 | 1   | 0                     | 3               | 0.982         | 1.096   | -1.468       | -0.765              | 1         |
| 1.03   | 0   | 1                     | 2               | -0.175        | -0.575  | 0.442        | -0.408              | 2         |
| -1.706 | 0   | 0                     | 0               | -1.203        | -1.077  | 1.198        | 1.38                | 0         |
| 1.656  | 1   | 2                     | 3               | 0.597         | 1.481   | -1.706       | -0.694              | 1         |
| -0.22  | 0   | 0                     | 2               | -0.561        | -0.726  | 0.76         | -0.194              | 2         |
| 0.483  | 1   | 1                     | 1               | 0.854         | 0.771   | -0.115       | -0.623              | 3         |

---

**Let's code!**

**[s.id/css26](#)**

# Feature-based models

This is a multi-class classification problem.

Target variable: **Diagnosis** = {Allergy, Common Cold, COVID19, Influenza}

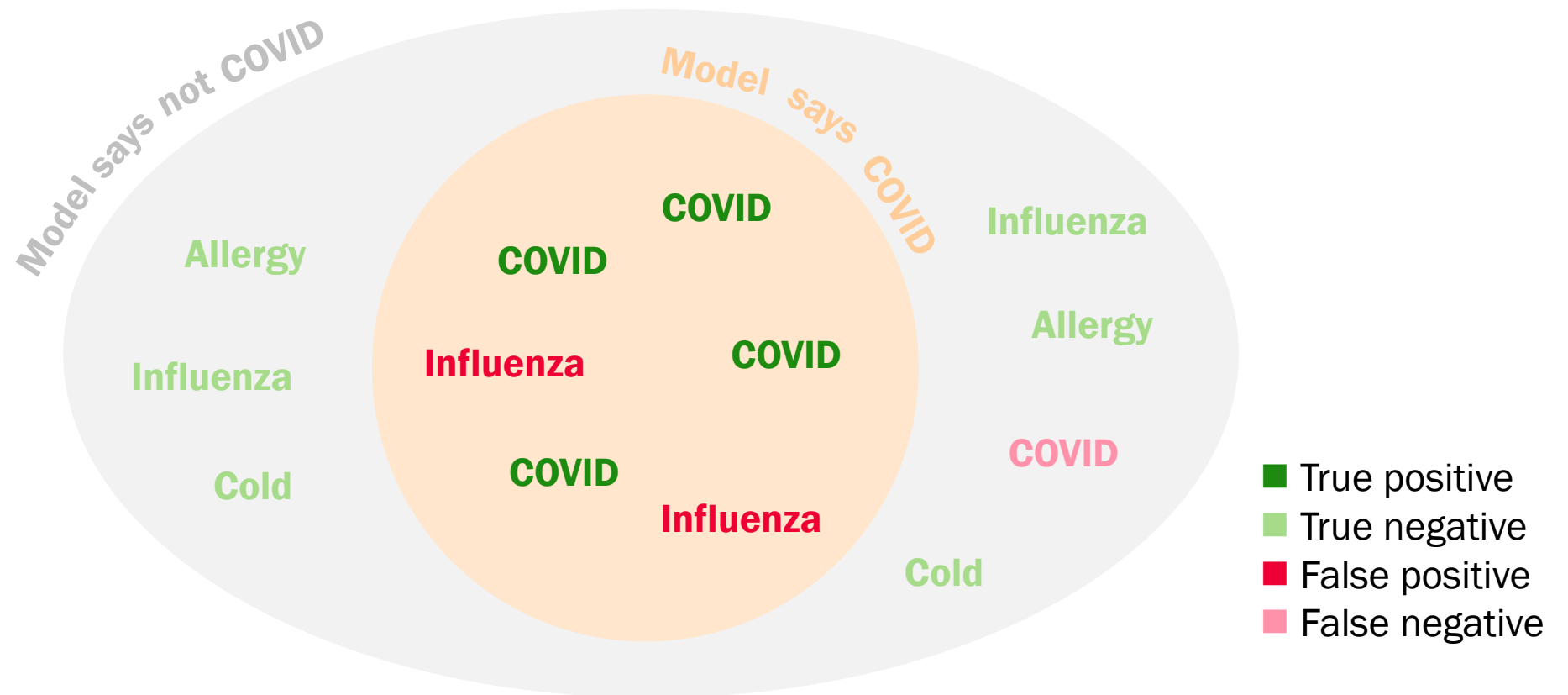
We can use different algorithms for training, e.g. Logistic Regression, Decision Trees, Random Forests, Neural Networks, Support Vector Machines, etc.

We train  $\Rightarrow$  (optimise)  $\Rightarrow$  test  $\Rightarrow$  use.

# Evaluation

## Classification

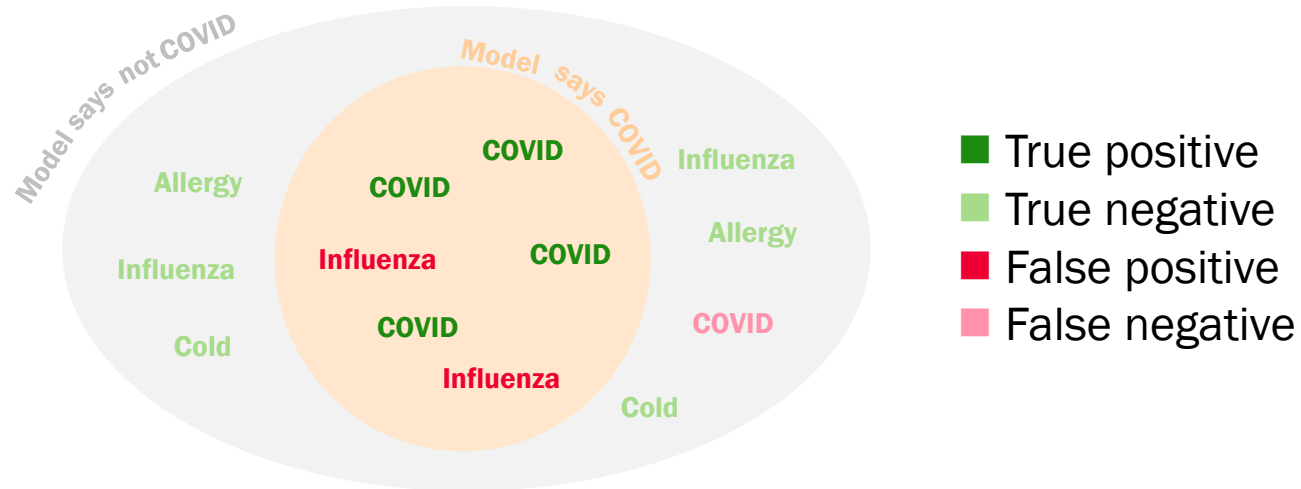
How do we know if our model is good? E.g. How well can we detect COVID?



# Evaluation

## Classification

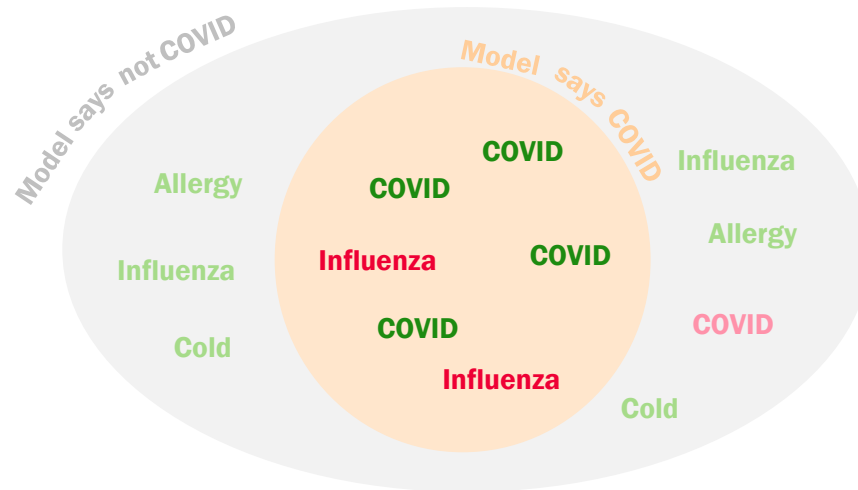
How well can we detect COVID?



# Evaluation

## Classification

How well can we detect COVID?



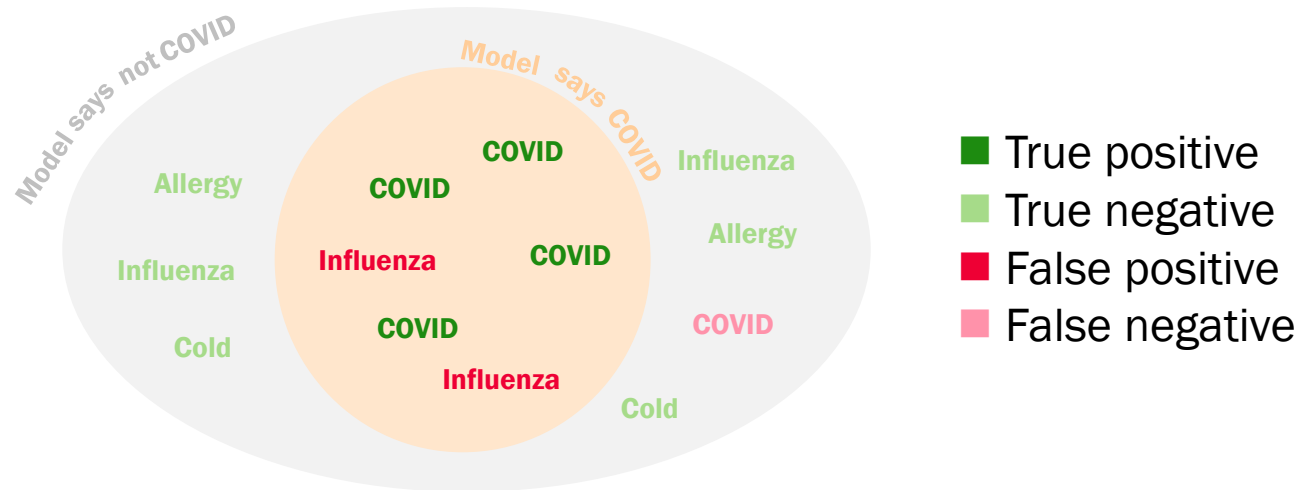
- True positive
- True negative
- False positive
- False negative

$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}}$$

# Evaluation

## Classification

How well can we detect COVID?

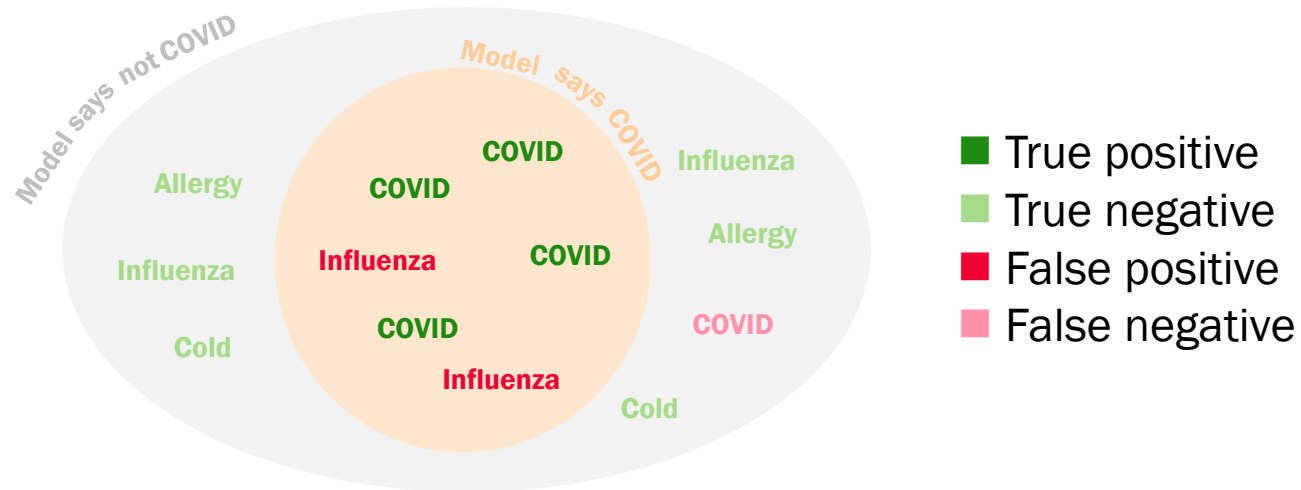


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# Evaluation

## Classification

How well can we detect COVID?



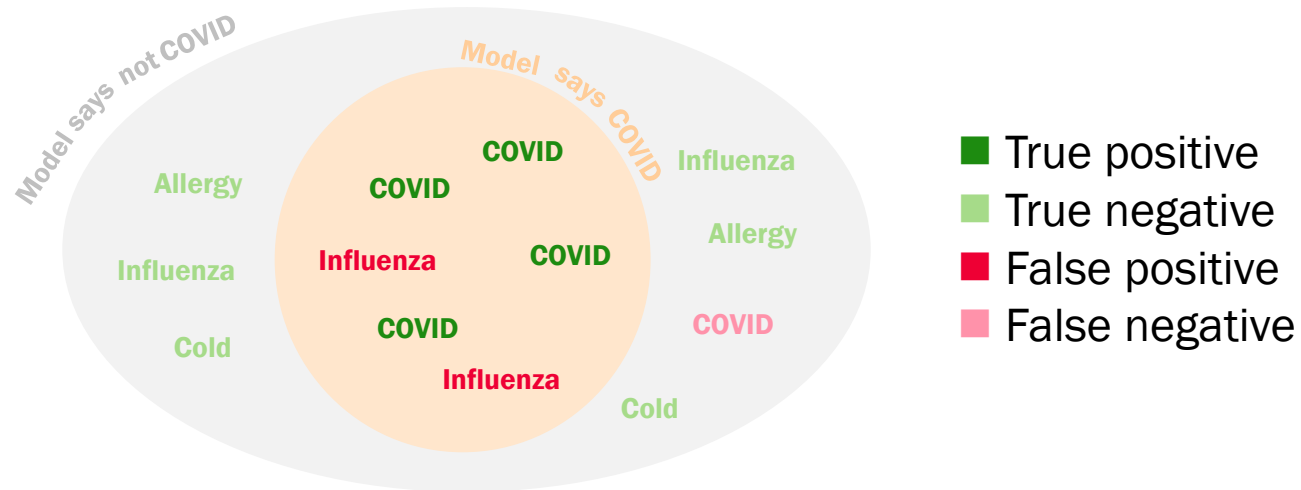
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$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}}$$

# Evaluation

## Classification

How well can we detect COVID?



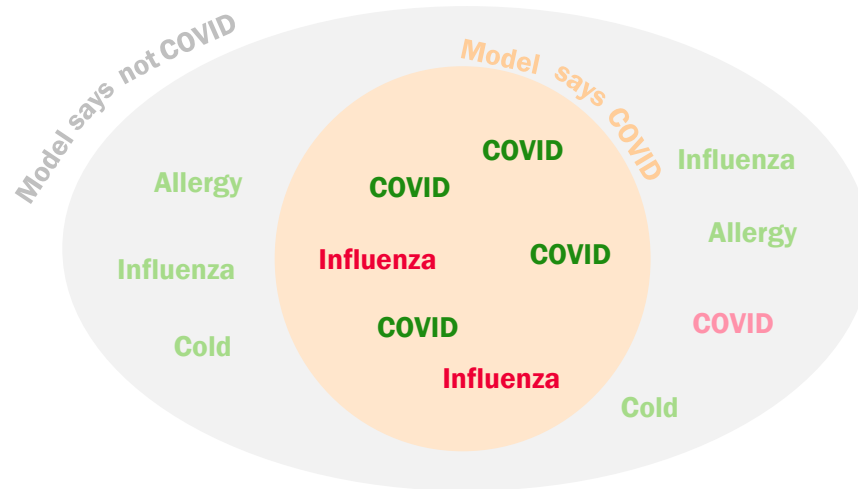
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$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} = \frac{4}{4 + 2} = \frac{4}{6} = \mathbf{0.67}$$

# Evaluation

## Classification

How well can we detect COVID?



- True positive
- True negative
- False positive
- False negative

$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}} = \frac{4 + 6}{13} = \frac{10}{13} = \mathbf{0.77}$$

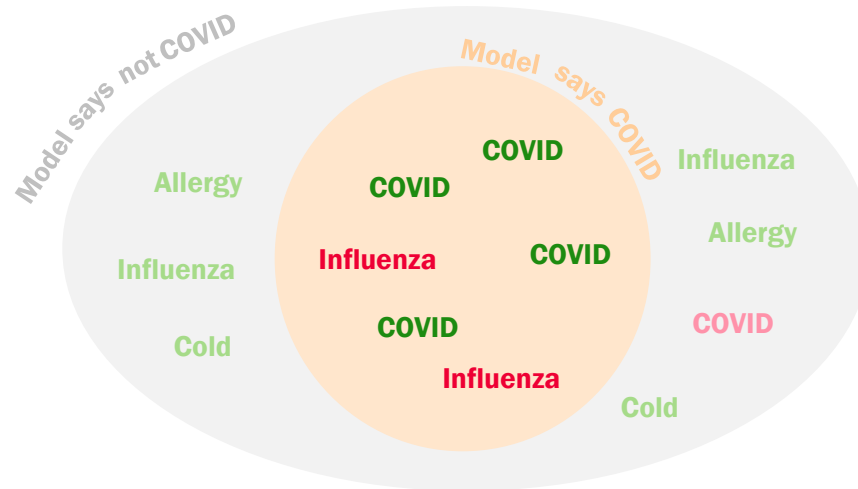
$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} = \frac{4}{4 + 2} = \frac{4}{6} = \mathbf{0.67}$$

$$\text{Recall} = \frac{\text{relevant and retrieved}}{\text{relevant}} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{True positive}}{\text{True positive} + \text{False negative}}$$

# Evaluation

## Classification

How well can we detect COVID?



$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}} = \frac{4 + 6}{13} = \frac{10}{13} = \mathbf{0.77}$$

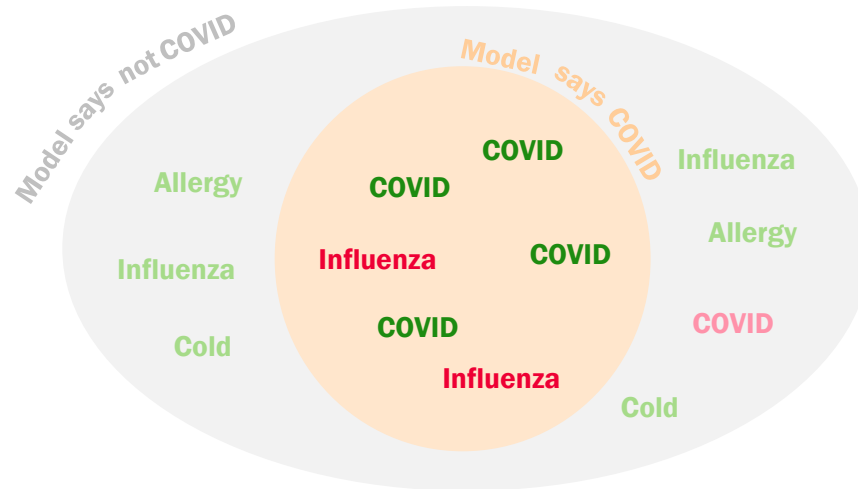
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$$\text{Recall} = \frac{\text{relevant and retrieved}}{\text{relevant}} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{True positive}}{\text{True positive} + \text{False negative}} = \frac{4}{4 + 1} = \frac{4}{5} = \mathbf{0.80}$$

# Evaluation

## Classification

How well can we detect COVID?



$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}} = \frac{4 + 6}{13} = \frac{10}{13} = \mathbf{0.77}$$

$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} = \frac{4}{4 + 2} = \frac{4}{6} = \mathbf{0.67}$$

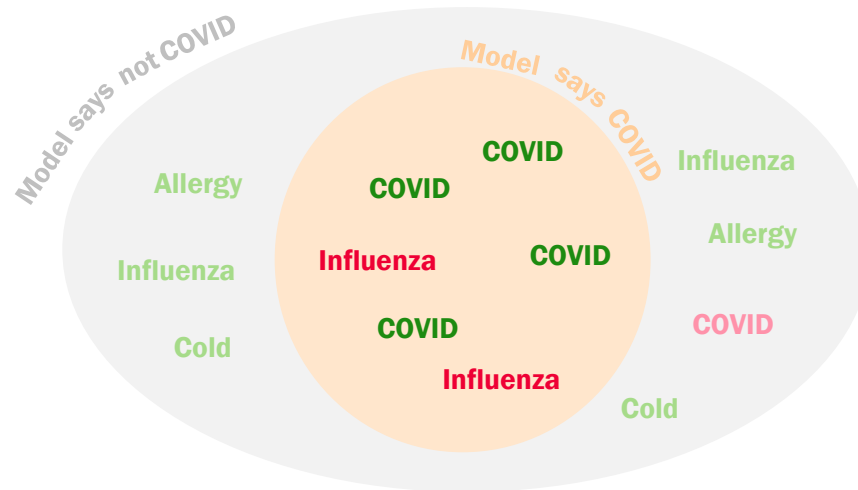
$$\text{Recall} = \frac{\text{relevant and retrieved}}{\text{relevant}} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{True positive}}{\text{True positive} + \text{False negative}} = \frac{4}{4 + 1} = \frac{4}{5} = \mathbf{0.80}$$

$$F = \frac{2 \times P \times R}{P + R}$$

# Evaluation

## Classification

How well can we detect COVID?



$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}} = \frac{4 + 6}{13} = \frac{10}{13} = \mathbf{0.77}$$

$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} = \frac{4}{4 + 2} = \frac{4}{6} = \mathbf{0.67}$$

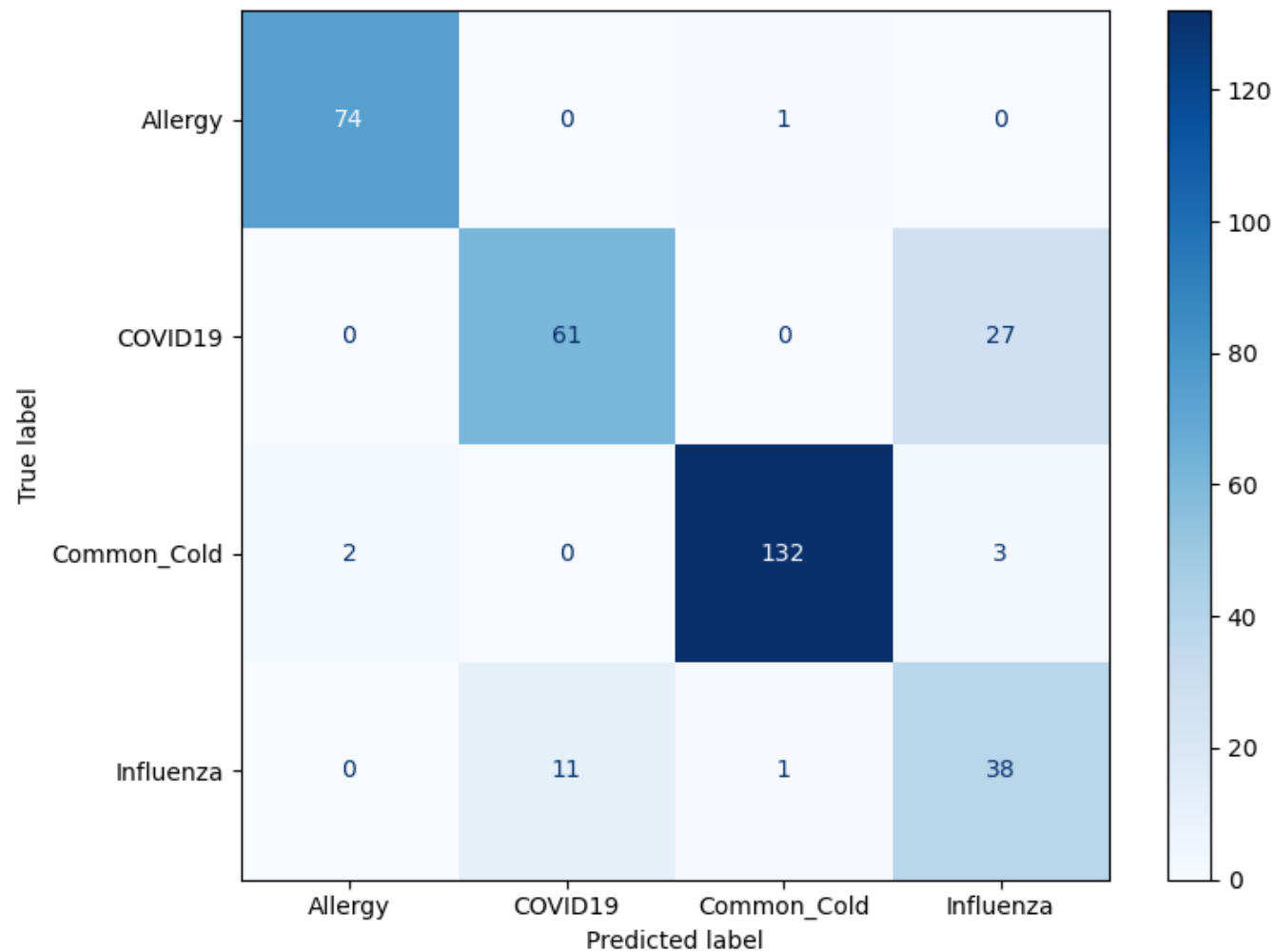
$$\text{Recall} = \frac{\text{relevant and retrieved}}{\text{relevant}} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{True positive}}{\text{True positive} + \text{False negative}} = \frac{4}{4 + 1} = \frac{4}{5} = \mathbf{0.80}$$

$$F = \frac{2 \times P \times R}{P + R} = \frac{2 \times 0.67 \times 0.80}{0.67 + 0.80} = \frac{1.072}{1.47} = \mathbf{0.73}$$

# Evaluation

## Classification

Confusion matrix



# Evaluation

## Regression

E.g. How accurately can we predict how many days will it take a patient to recover from COVID?

| Patient | Actual ( $y_i$ ) | Predicted ( $\hat{y}_i$ ) |
|---------|------------------|---------------------------|
| 1       | 10               | 12                        |
| 2       | 14               | 13                        |
| 3       | 7                | 8                         |
| 4       | 21               | 15                        |
| 5       | 18               | 17                        |

$$\text{Mean Average Error (MAE)} = \frac{1}{n} \sum |y_i - \hat{y}_i| = \frac{2 + 1 + 1 + 6 + 1}{5} = \frac{11}{5} = \mathbf{2.20 \text{ days}}$$

$$\text{Root Mean Squared Error (RMSE)} = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} = \sqrt{\frac{4 + 1 + 1 + 36 + 1}{5}} = \sqrt{\frac{43}{5}} = \sqrt{8.60} \approx \mathbf{2.93 \text{ days}}$$

$$\text{Coefficient of Determination (R2)} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2} = \frac{43}{130} = 1 - 0.3308 = \mathbf{0.67}$$

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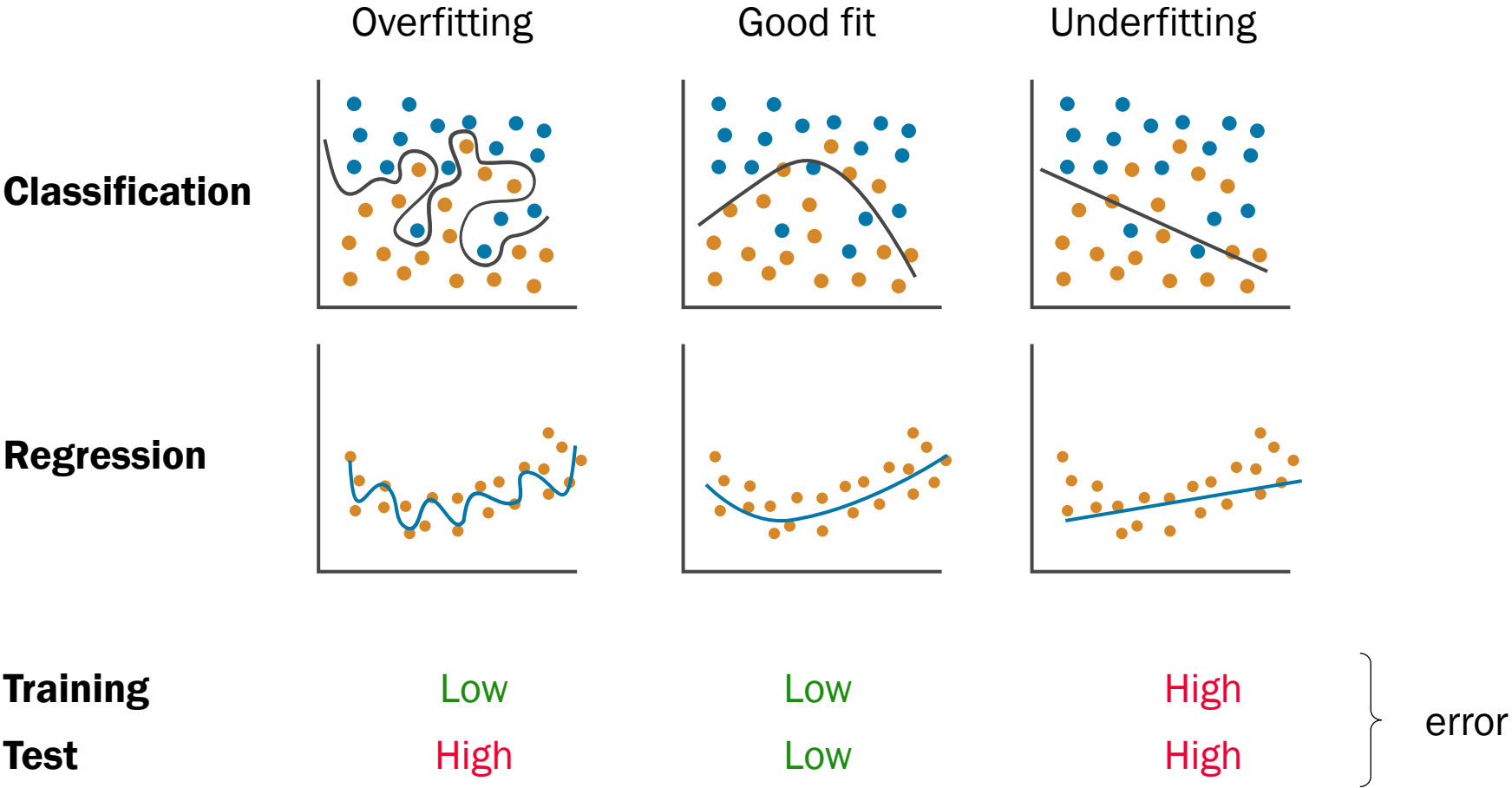
# Evaluation

## Specialist metrics

- **Automatic Speech Recognition (ASR)  $\Rightarrow$  Word Error Rate (WER)**  
How many word mistakes the speech-to-text model makes.
- **Machine Translation (MT)  $\Rightarrow$  BLEU**  
How closely a computer translation matches a human translation.
- **Text Summarisation  $\Rightarrow$  ROUGE**  
How much key information from the original text was kept in the summary.
- **Grammatical Error Correction (GEC)  $\Rightarrow$   $F_{0.5}$ -Score**  
Correction quality, heavily favoring precision over recall.
- **Search & Recommendation Engines  $\Rightarrow$  Mean Reciprocal Rank / Normalized Discounted Cumulative Gain (NDCG)**  
Whether the best answers appear at the top of the search list.



# Overfitting and underfitting





**See you on Friday!**

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